2

Standard DEB model in time, length and energy

This chapter discusses the simplest non-degenerated DEB model that is implied by DEB theory, the standard or canonical DEB model, to show the concepts of the previous chapter in action. The next chapter introduces more concepts on chemical transformations to deal with more complex situations. The standard DEB model assumes isomorphy and has a single reserve and a single structure, which is appropriate for many aspects of the metabolic performance of animals; other organisms typically require more reserves, and some (e.g. plants) also more structures. So in this chapter, we keep an animal in mind as a reference, which helps to simplify the phrasing. In this chapter substrate (food) has a constant composition that matches the needs of the individual. Food density in the environment might vary in the standard DEB model, but the discussion of what happens during prolonged starvation is delayed to Chapter 4. The discussion of mass aspects is also delayed and we here only use time t, length L and energy E. Length L is the structural (volumetric) length, $L = V^{1/3}$, where V is the structural volume. We use energy only conceptually, and typically in scaled form, and also delay the discussion of its quantification. The discussion of energy aspects does not imply that the individual should be energy-limited.

The logic of the energy flows will be discussed in this chapter and we start with a brief overview in this introductory section. We here keep the amount of detail to a minimum, neglecting all fast process, and focusing on the slow ones that matter on a life-cycle basis. Since reserve dynamics is slow relative to gut content dynamics, see {256}, the latter is not part of the standard DEB model. The dynamics of blood composition is linked to the dynamics of gut contents, so we neglect the blood compartment as well. Most aspects of behaviour are even faster than the dynamics of gut contents, so behaviour is here treated in very much reduced form.

The relationships between the different processes are schematically summarised in Figure 2.1. Food is ingested by a post-embryonic animal, transformed into faeces and egested; defecation is a special case of product formation. The feeding rate depends on food availability and the amount of structure. Energy, in the form of metabolites, is derived from food and added to the reserve. The reserve is mobilised at a rate that depends on the

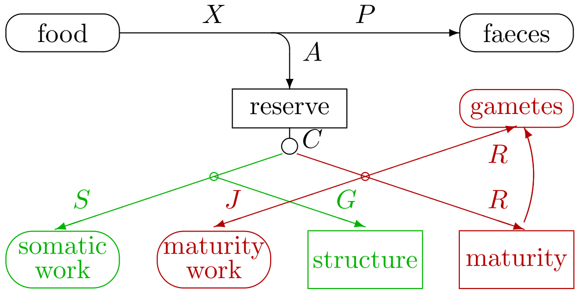


Figure 2.1: Energy fluxes in the standard DEB model. The rounded boxes indicate sources or sinks. The symbols stand for: X food intake; P defecation; A assimilation; C mobilisation; S somatic maintenance; J maturity maintenance; G growth; R reproduction. See Figure 10.4 for an evolutionary setting of this diagram.

amount of reserve and the amount of structure, and used for metabolic purposes. A fraction κ of the mobilised reserve flux is used for somatic maintenance plus growth, the rest for maturity maintenance plus maturation (in juveniles) or reproduction (in adults). Somatic maintenance has priority over growth, so growth ceases if all energy available for maintenance plus growth is used for somatic maintenance. Likewise maturity maintenance has priority over maturation or reproduction. Reserve that is allocated to reproduction is first collected in a buffer; the reproduction buffer has species-specific handling rules for transformation into eggs that typically leave the body upon formation. Eggs consist initially almost exclusively of reserve, the amount of structure, and the level of maturity being negligibly small. The reserve density at birth (hatching) equals that of the mother at egg formation, a maternal effect.

Each of these processes will be quantified in the following sections on the basis of a set of simple assumptions that are collected in Table 2.4.

2.1 Feeding

Feeding is part of the behavioural repertoire and, therefore, notoriously erratic compared with other processes involved in energetics. The three main factors that determine feeding rates are food availability, body size and temperature.

2.1.1 Food availability is per volume or surface area of environment

For some species it is sensible to express food availability per surface area of environment, for others food per volume makes more sense, and intermediates also exist. The body size of the organism and spatial heterogeneity of the environment hold the keys to the classification. Food availability for krill, which feed on algae, is best expressed in terms of biomass or biovolume per volume of water, because this links up with processes that determine filtering rates. The spatial scale at which algal densities differ is large with respect to the body size of the krill. Baleen whales, which feed on krill, are intermediate between surface and volume feeders because some dive below the top layer, where most algae and krill are located, and sweep the entire column to the surface; so it does not matter where the krill is in the column. Cows and lions are typically surface feeders and food availability is most appropriately expressed in terms of biomass per surface area.

To avoid notational complexity, we here express food density X relative to the value that results in half of the maximum food intake rate, K, and treat $x = X/K$ as a dimensionless quantity. K is known as the half-saturation coefficient or Michaelis–Menten constant.

2.1.2 Food transport is across surface area of individual

Organisms use many methods to obtain their meal; some sit and wait for the food to pass by, others search actively. Figure 2.2 illustrates a small sample of methods, roughly classified with respect to active movements by prey and predator. The food items can be very small with respect to the body size of the individual and rather evenly distributed over the environment, or the food can occur in a few big chunks. This section briefly mentions some feeding strategies and explains why feeding rates tend to be proportional to the surface area when a small individual is compared to a large one of the same species. (Comparisons between species are made in Chapter 8, {287}.) The examples illustrate a simple physical principle: mass transport from one environment to another, namely to the organism, must be across a surface, so the ingestion rate in watts is

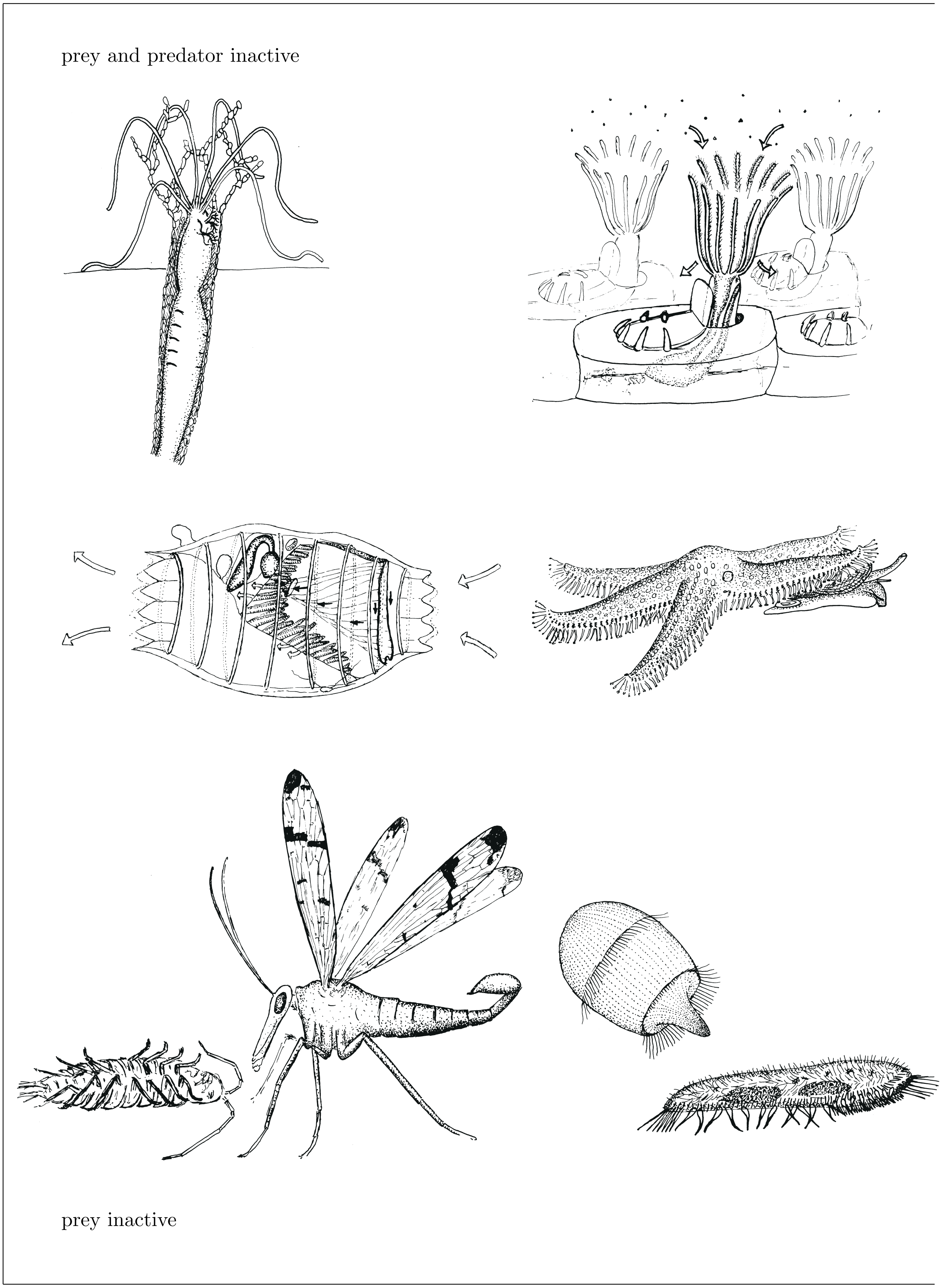
$$\dot{p}_X = \{\dot{p}_X\} L^2 \tag{2.1}$$

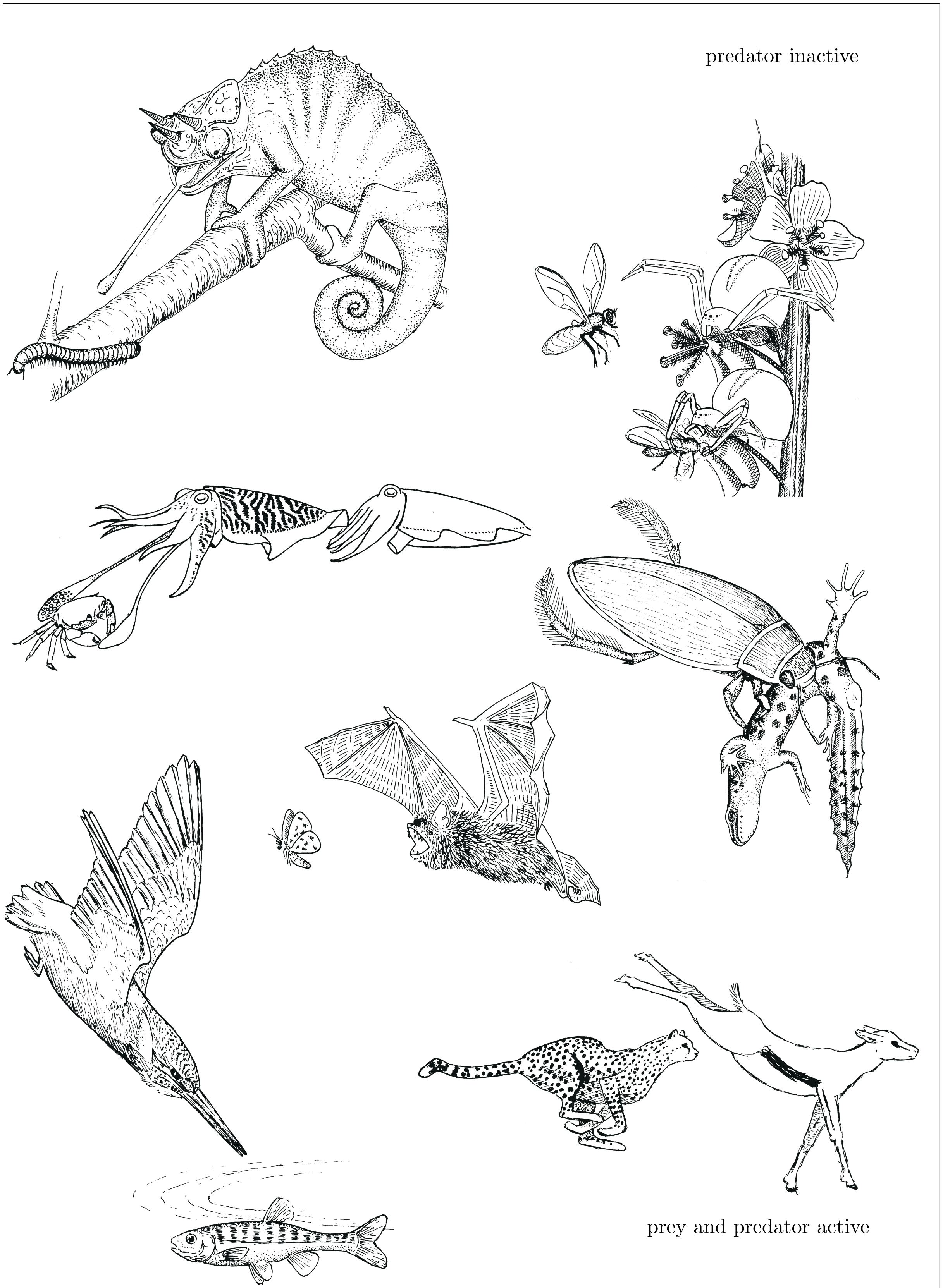
where L is the structural (volumetric) length of the individual, and the specific ingestion rate $\{\dot{p}_X\}$ is some function of food density. Notice that not all of the surface of an isomorph needs to be involved in food acquisition, it might be a fixed fraction of it, such as gut surface area.

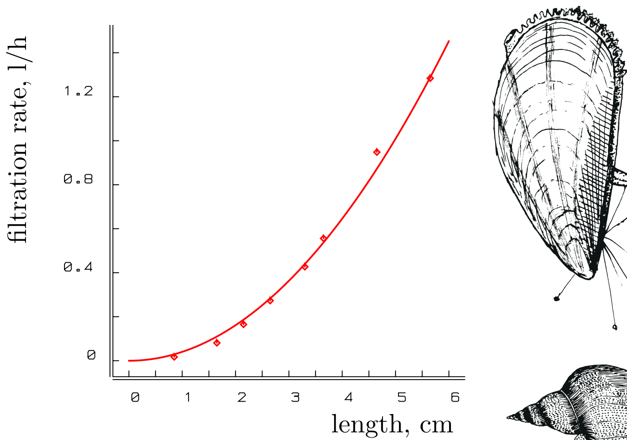
Marine polychaetes, sea anemones, sea lilies and other species that feed on blind prey are rather apathetic. Sea lilies simply orient their arms perpendicular to an existing current (if mild) at an exposed edge of a reef and take small zooplankters by grasping them one by one with many tiny feet. The arms form a rather closed fan in mild currents, so the active area is proportional to the surface area of the animal. Sea gooseberries stick plankters to the side branches of their two tentacles using cells that are among the most complex in the animal kingdom. Since the length of the side branches as well as the tentacles are proportional to the length of the animal, the encounter probability is proportional to a surface area.

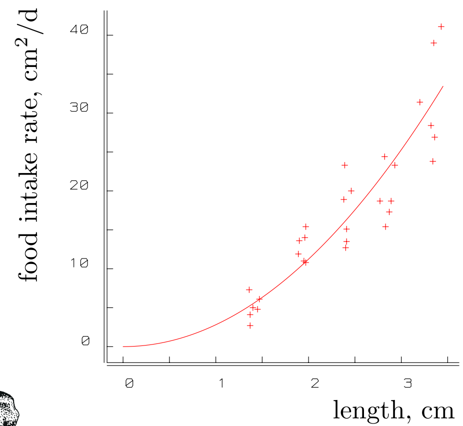
Filter feeders, such as daphnids, copepods and larvaceans, generate water currents of a strength that is proportional to their surface area [158], because the flapping frequency of their limbs or tails is about the same for small and large individuals [910], and the current is proportional to the surface area of these extremities. (Allometric regressions of currents give a proportionality with length to the power 1.74 [147], or 1.77 [312] in daphnids. In view of the scatter, they agree well with a proportionality with squared length.) The ingestion rate is proportional to the current, so to squared length. Allometric regressions of ingestion rates resulted in a proportionality with length to the power 2.2 [770], 1 [911], 2.4–3 [262] and 2.4 [862] in daphnids. This wide range of values illustrates the limited

Figure 2.2: A small sample of feeding methods classified with respect to the moving activities of prey and predator.


prey and predator inactive
prey inactive


predator inactive
prey and predator active





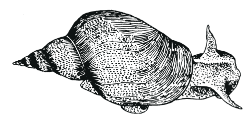


Figure 2.3: Filtration rate as a function of shell length, L, of the blue mussel *Mytilus edulis* at constant food density (40×10^6 cells l^{-1} *Dunaliella marina*) at 12 °C. Data from Winter [1258]. The least-squares-fitted curve is $\{\dot{F}\}L^2$, with $\{\dot{F}\} = 0.041\,\mathrm{l\,h^{-1}\,cm^{-2}}$.

Figure 2.4: Lettuce intake as a function of shell length, L, in the pond snail *Lymnaea stagnalis* at 20 °C [1292]. The weighted least-squares-fitted curve is $\{\dot{J}_{XA}\}L^2$, with $\{\dot{J}_{XA}\} = 2.81\,\mathrm{cm^2\,d^{-1}\,cm^{-2}}$.

degree of replicatability of these types of measurements. This is partly due to the inherent variability of the feeding process, and partly to the technical complications of measurement. Feeding rate depends on food density, as is discussed on {32}, while most measurement methods make use of changes in food densities so that the feeding rate changes during measurement. Figure 2.10 illustrates results obtained with an advanced technique that circumvents this problem [330].

The details of the filtering process differ from group to group. Larvaceans are filterers in the strict sense; they remove the big particles first with a coarse filter and collect the small ones with a fine mesh, see Figure 2.19. The collected particles are transported to the mouth in a mucous stream generated by a special organ, the endostyle. Copepods take their minute food particles out of the water, one by one with grasping movements [1180]. Daphnids exploit centrifugal force and collect them in a groove; Figure 2.10 shows that the resulting feeding rate is proportional to squared length. Ciliates, bryozoans, brachiopods, bivalves and ascidians generate currents, not by flapping extremities but by beating cilia on part of their surface area. The ciliated part is a fixed portion of the total surface area [366], and this again results in a filtering rate proportional to squared length; see Figure 2.3.

Some surface feeding animals, such as crab spiders, trapdoor spiders, praying mantis, scorpion fish and frogs, lie in ambush; their prey will be snatched upon arrival within reach, i.e. within a distance that is proportional to the length of a leg, jaw or tongue. The catching probability is proportional to the surface area of the predatory isomorphs. When aiming at a prey with rather keen eye sight, they must hide or apply camouflage.

Many animals search actively for their meal, be it plant or animal, dead or alive. The standard cruising rate of surface feeders tends to be proportional to their length,

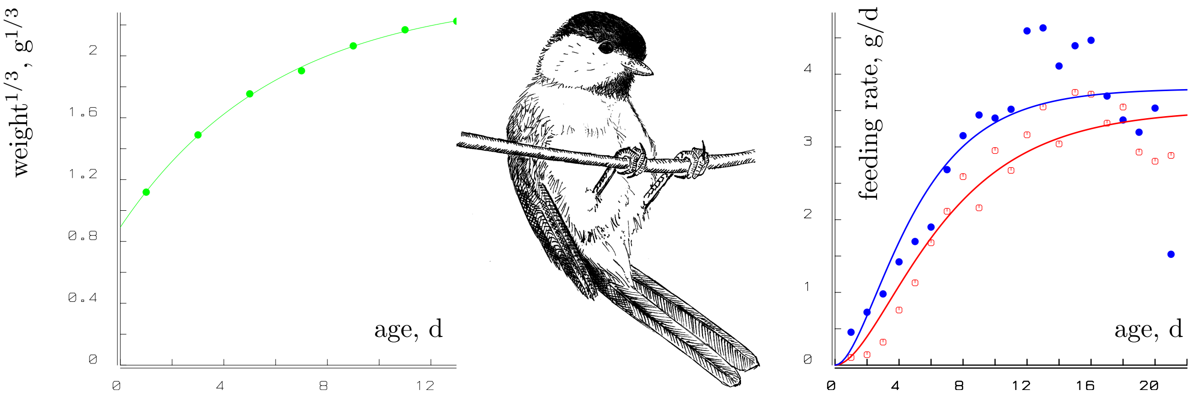


Figure 2.5: The von Bertalanffy growth curve applies to the black-capped chickadee, *Parus atricapillus* (left figure, data from Kluyver [606, 1080]. Brood size was a modest 5.) The amount of food fed per male (•) or female (∘) nestling in the closely related mountain chickadee, *P. gambeli*, is proportional to weight$^{2/3}$ (right figure), as might be expected for individuals that grow in a von Bertalanffy way. Data from Grundel [438, 1080]. The last five data points were not included in the fit; the parents stop feeding, and the young still have to learn to gather food while rapidly losing weight.

because the energy investment in movement as part of the maintenance costs tends to be proportional to volume, while the energy costs of transport are proportional to surface area; see {31}. Proportionality of cruising rate to length also occurs if limb movement frequency is more or less constant [921]. The width of the path searched for food by cows or snails is proportional to length if head movements perpendicular to the walking direction scale isomorphically. So feeding rate is again proportional to surface area, which is illustrated in Figure 2.4 for the pond snail.

The duration of a dive for the sperm whale *Physeter macrocephalus*, which primarily feeds on squid, is proportional to its length, as is well known to whalers [1219]. This can be understood, since the respiration rate of this endotherm is approximately proportional to surface area, as I argue on {143}, and the amount of reserve dioxygen is proportional to volume on the basis of a homeostasis argument. It is not really obvious how this translates into the feeding rate, if at all; large individuals tend to feed on large prey, which occur less frequently than small prey. Moreover, time investment in hunting can depend on size as well. If the daily swimming distance during hunting were independent of size, the searched water volume would be about proportional to surface area for a volume feeder such as the sperm whale. If the total volume of squid per volume of water is about constant, this would imply that feeding rate is about proportional to surface area.

The amount of food parent birds feed per nestling relates to the requirements of the nestling, which is proportional to surface area; Figure 2.5 illustrates this for chickadees. This is only possible if the nestlings can make their needs clear to the parents, by crying louder: demand systems in the strict sense of the word.

Catching devices, such as spider or pteropod webs and larvacean filter houses [19], have effective surface areas that are proportional to the surface area of the owner.

Bacteria, floating freely in water, are transported even by the smallest current, which implies that the current relative to the cell wall is effectively nil. Thus bacteria must obtain substrates through diffusion, {259}, or attach to hard surfaces (films) or each other (flocs, {132}) to profit from convection, which can be a much faster process. Some species develop more flagellae at low substrate densities, which probably reduces diffusion limitation (L. Dijkhuizen, pers. comm.). Uptake rate is directly proportional to surface area, if the carriers that bind substrate and transport it into the cell have a constant frequency per unit surface area of the cell membrane [8, 177]. *Arthrobacter* changes from a rod shape into a small coccus at low substrate densities to improve its surface area/volume ratio. Caulobacters do the same by enhancing the development of stalks under those conditions [903].

Some fungi, slime moulds and bacteria glide over or through the substrate, releasing enzymes and collecting elementary compounds via diffusion. Upon arrival at the cell surface, the compounds are taken up actively. The bakers' yeast *Saccharomyces cerevisiae* typically lives as a free floating, budding unicellular, but under nitrogen starvation it can switch to a filamentous multicellular phase, which can penetrate solids [518]. Many protozoans engulf particles (a process known as phagocytosis) with their outer membrane (again a surface), encapsulate them into a feeding vacuole and digest them via fusion with bodies that contain enzymes (lysosomes). Such organisms are usually also able to take up dissolved organic material, which is much easier to quantify. In giant cells, such as the Antarctic foraminiferan *Notodendrodes*, the uptake rate can be measured directly and is found to be proportional to surface area [257]. Ciliates use a specialised part of their surface for feeding, which is called the cytostome; isomorphic growth here makes feeding rate proportional to surface area again.

All these different feeding processes relate to surface areas in comparisons between different body sizes within a species at a constant low food density. At high food densities, the encounter probabilities are no longer rate–limiting, this becomes the domain of digestion and other food processing activities involving other surface areas, for example the mouth opening and the gut wall. The gradual switch in the leading processes becomes apparent in the functional response, i.e. the ingestion rate as function of food density, {32}.

2.1.3 Feeding costs are paid from food directly

As feeding methods are rather species-specific, costs of feeding will also be species-specific if they contribute substantially to the energy budget. I argue here that costs of feeding and movements that are part of the routine repertoire are usually insignificant with respect to the total energy budget. For this reason this subsection does not do justice to the voluminous amount of work that has been done on the energetics of movements [877], a field that is of considerable interest in other contexts. Alexander [14] gives a most readable and entertaining introduction to the subject of energetics and biomechanics of animal movement. Differences in respiration between active and non-active individuals give a measure for the energy costs of activity, but metabolic activity might be linked to behavioural activity more generally, which complicates the interpretation. The resting metabolic rate is a measure that excludes active movement. The standard or basal metabolic rate includes a low level of movement only. The field metabolic rate is the daily energy expenditure for free–ranging individuals. Karasov [575] found that the field metabolic rate is about twice the standard metabolic rate for several species of mammal, and that the costs of locomotion ranges 2–15 % of the field metabolic rate. Mammals are among the more active species. The respiration rate associated with filtering in animals such as larvaceans and ascidians was found to be less than 2 % of the total dioxygen consumption [356]. Energy investment in feeding is generally small, which does not encourage the introduction of many parameters to describe this investment. Feeding costs can be accommodated in two ways within the DEB theory without the introduction of new parameters, and this subsection aims to explore to what extent this accommodation is realistic.

The first way to accommodate feeding costs is when they are proportional to feeding rate. They then show up as a reduction of the energy gain per unit of food. One can, however, argue that feeding costs per unit of food should increase with decreasing food density, because of the increased effort of extracting it from the environment. This type of cost can only be accommodated without complicating the model structure if these costs cancel against increased digestion efficiency, caused by the increased gut residence time, see {266}.

The second way to accommodate feeding costs without complicating the model structure applies if the feeding costs are independent of the feeding rate and proportional to body volume. They then show up as part of the maintenance costs, see {42}. This argument can be used to understand how feeding rates for some species tend to be proportional to surface area if transportation costs are also proportional to surface area, so that the cruising rate is proportional to length, {28}. In this case feeding costs can be combined with costs of other types of movement that are part of the routine repertoire. A fixed (but generally small) fraction of the maintenance costs then relates to movement.

Schmidt-Nielsen [1020] calculated $0.65\,\mathrm{ml\,O_2\,cm^{-2}\,km^{-1}}$ to be the surface-area-specific transportation costs for swimming salmon, on the basis of Brett's work [148]. (He found that transportation costs are proportional to weight to the power 0.746, but respiration was not linear with speed. No check was made for anaerobic metabolism of the salmon. Schmidt-Nielsen obtained, for a variety of fish, a power of 0.7, but 0.67 also fits well.) Fedak and Seeherman [339] found that the surface-area-specific transportation cost for walking birds, mammals and lizards is about $5.39\,\mathrm{ml\,O_2\,cm^{-2}\,km^{-1}} \simeq 118\,\mathrm{J\,cm^{-2}\,km^{-1}}$. (They actually report that the transportation costs are proportional to weight to the power 0.72 as the best fitting allometric relationship, but the scatter is such that 0.67 fits as well.) This is consistent with data from Taylor *et al.* [1135] and implies that the costs of swimming are some 12 % of the costs of running. Their data also indicate that the costs of flying are between those of swimming and running and amount to some $1.87\,\mathrm{ml\ O_2\,cm^{-2}\,km^{-1}}$.

The energy costs of swimming are frequently taken to be proportional to squared speed and squared length on sound mechanical grounds [671], which questions the usefulness of the above-mentioned costs and comparisons because the costs of transportation become dependent on speed. If the inter-species relationship that speeds scale with the square root of volumetric length, see {302}, also applies to intra-species comparisons, the transportation costs are proportional to volume if the travelling time is independent of size.

The energy required for walking and running is found to be proportional to velocity

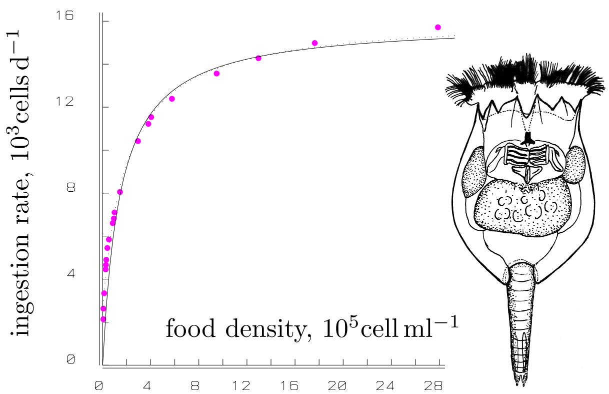


Figure 2.6: The ingestion rate, $\dot{p}_X$, of an individual (female) rotifer *Brachionus rubens*, feeding on the green alga *Chlorella* at 20 °C, as a function of food density, X. Data from Pilarska [897]. The curve is the hyperbola (2.2), with maximum feeding rate $15.97\,10^3$ cells d^{-1} and saturation coefficient $K = 1.47\,10^5$ cells ml^{-1}. The stippled curve allows for an additive error in the measurement of the algal density of $0.35\,10^5$ cells ml^{-1}.

for a wide diversity of terrestrial animals including mammals, birds, lizards, amphibians, crustaceans and insects [387]. This means that the energy costs of walking or running a certain distance are independent of speed and just proportional to distance. If the costs of covering a certain distance are dependent on speed, and temperature affects speeds, these costs would work out in a really complex way at the population and community levels.

The conclusion is that, for the purposes of studying how energy budgets change during the life span, transportation costs either show up as a reduction of energy gain from food, or as a fixed fraction of the somatic maintenance costs when these costs are proportional to structural mass.

2.1.4 Functional response converts food availability to ingestion rate

The feeding or ingestion rate, $\dot{p}_X$, of an organism as a function of scaled food density, $x = X/K$, expressed in watts, is described well by the hyperbolic functional response

$$\dot{p}_X = f\dot{p}_{Xm} = f\{\dot{p}_{Xm}\}L^2 \quad \text{with } f = \frac{x}{1+x} \tag{2.2}$$

where $\dot{p}_{Xm}$ the maximum ingestion rate, $\{\dot{p}_{Xm}\}$ the specific maximum ingestion rate and L the volumetric length. The Michaelis–Menten function f is referred to as the scaled functional response. Holling [522] called it type II functional response, and it is illustrated in Figure 2.6. It applies to the intake of organic particles by ciliates (phagocytosis), the filtering of algae by daphnids, the catching of flies by mantis, the uptake of substrates by bacteria, the uptake of nutrients by algae and plants, and the transformation of substrates by enzymes. Although these processes differ considerably in detail, some common abstract principle gives rise to the hyperbolic functional response: the busy period, which is characteristic of the Synthesising Unit, see {101}.

All behaviour is classified into just two categories: food acquisition and food processing, which not only includes food handling, but also digestion and other metabolic steps that keep the individual away from food acquisition. These two behavioural components compete for time allocation by the individual.

Let $\dot{F}$ denote the filtering rate (in volume per time), a rate that is taken to depend on mean particle density only, and not on particle density at a particular moment. The arrival rate of food particles (in number per time), present in density N (in number per volume), equals $\dot{h} = \dot{F}N$. Notice that we have to multiply N with the mass per particle M_X to arrive at the mass of food per volume of environment $X = NM_X$, and that we have to multiply $\dot{h}$ with M_X to arrive at the ingestion rate $\dot{J}_{XA} = M_X\dot{h}$ in terms of C-moles per time. The mean time between the end of a handling period and the next arrival (the binding period) is $t_b = (N\dot{F})^{-1}$. The mean handling period is $t_p = \dot{h}_m^{-1}$, which is the maximum value of $\dot{h}$, where no time is lost in waiting for particles. The time required to find and eat one particle is thus given by $t_c = t_b + t_p$ and the mean ingestion rate is $\dot{h} = t_c^{-1} = \dot{h}_m N(\dot{h}_m/\dot{F} + N)^{-1}$, which is hyperbolic in the density X. The (half) saturation coefficient is inversely proportional to the product of the handling time and the filtering (or searching) rate, i.e. $K = M_X(t_p\dot{F})^{-1} = \dot{J}_{XAm}/\dot{F}$, where $\dot{J}_{XAm}$ is the maximum food intake rate in moles per time.

Filter feeders, such as rotifers, daphnids and mussels, reduce filtering rate with increasing food density [365, 910, 970, 971], rather than maintain a constant rate, which would imply the rejection of some food particles. They reduce the rate by such an amount that no rejection occurs because of the handling (processing) of particles. If all incoming water is swept clear, the filtering rate is found from $\dot{F}(X) = \dot{h}/N$, which reaches a maximum if no food is around (temporarily), so that $\dot{F}_m = \{\dot{h}_m\}L^2/N$, and $\dot{F}$ approaches zero for high food densities. An alternative interpretation of the saturation coefficient in this case would be $K = \dot{J}_{XAm}/\dot{F}_m = \{\dot{J}_{XAm}\}/\{\dot{F}_m\}$, which is independent of the size of the animal, as long as only intra-specific comparisons are made. It combines the maximum capacity for food searching behaviour, only relevant at low food densities, with the maximum capacity for food processing, which is only relevant at high food densities. This mechanistic interpretion of the saturation coefficient is a special case of the dynamics of Synthesizing Units, which will be discussed in Chapter 3, {97}.

Because the specific searching rate $\{\dot{F}_m\}$ is closer to the underlying feeding process, it will be treated as primary parameter, rather than the descriptive saturation coefficient $K = \{\dot{J}_{XAm}\}/\{\dot{F}_m\}$.

A most interesting property of the hyperbolic functional response is that it is the only one with a finite number of parameters that maps onto itself. For instance, an exponential function of an exponential function is not again an exponential function. A polynomial (of degree higher than one) of a polynomial is also a polynomial, but it is of an increasingly higher degree if the mapping is repeated over and over again. The ratio of two linear functions of a ratio of two linear functions as in (2.2) is, however, again such a ratio; the linear function is a special case of this. In a metabolic pathway each product serves as a substrate for the next step. Neither the cell nor the modeller needs to know the exact number of intermediate steps to relate the production rate to the original substrate density, if and only if the functional responses of the subsequent intermediate steps are of the hyperbolic type. If, during evolution, an extra step is inserted in a metabolic pathway the performance of the whole chain does not change in functional form. This is a crucial point because each pathway has to be integrated with other pathways to ensure the proper functioning of the individual as a whole. If an insert in a metabolic pathway simultaneously required a qualitative change in regulation at a higher level, the probability of its occurrence during the evolutionary process would be remote. This suggests that complex regulation systems in metabolic pathways fix and optimise the kinetics that originate from the simpler kinetics on which Synthesising Units are based.

A most useful property of the hyperbolic functional response is that it has only two parameters that serve as simple scaling factors on the food density and ingestion rate axis. So if food density is expressed in terms of the saturation coefficient, and ingestion rate in terms of maximum ingestion rate, the functional response no longer has dimensions or parameters.

When starved animals are fed, they often ingest at a higher rate for a short time [1220], but this is here neglected. Starved daphnids, for instance, are able to fill their guts within 7.5 minutes [392].

2.1.5 Generalisations: differences in size of food particles

The derivation of the functional response can be generalised in different ways without changing the model. Each arriving particle can have an attribute that stands for its probability of becoming caught. The ith particle has some fixed probability ρ_i of being caught upon encountering an animal if the animal is not busy handling particles, and a probability of 0 if it is. The mass of each particle does not need to be the same. The flux $\dot{J}_X$ should be interpreted as the mean mass flux (in C-moles per time), where the mass of each particle represents a random trial from some frequency distribution.

It is not essential for the handling time to be the same for all particles; handling time can be conceived of as a second attribute attached to each particle, but it must be independent of food density. The amount of time required for food processing is taken to be proportional to the amount of mass of the food item to ensure that the maximum uptake capacity is not exceeded.

The condition of zero catching probability when the animal is busy can be relaxed. Metz and van Batenburg [782, 783] and Heijmans [481] tied catching probability to satiation (thought to be related to gut content in the mantis). An essential condition for hyperbolic functional responses is that catching probability equals zero if satiation (gut content) is maximal.

When offered different food items, individuals can select for size. Shelbourne [1044] reports that the mean length of *Oikopleura* eaten by plaice larvae increases with the size of the larvae. Copepods appear to select the larger algal cells [1123]. Daphnids do not collect very small particles, $< 0.9\,\mu$m cross-section [420], or large ones, > 27 and $> 71\,\mu$m; the latter values were measured for daphnids of length 1 and 3 mm respectively [175]. Kersting and Holterman [592] found no size selectivity between 15 and $105\,\mu\text{m}^3$ (and probably $165\,\mu\text{m}^3$) for daphnids. Size selection is rarely found in daphnids [968], or in mussels [365, 1248], but selection of food type does occur frequently [132].

Deviations from the hyperbolic functional response can be expected if the mass per particle is large, while the intensity of the arrival process is small. They can also result from, e.g. more behavioural traits, see {256}, social interactions, see {257}, transport processes of resources, see {259}.

2.2 Assimilation

The term 'assimilated' energy here denotes the free energy fixed into reserves; it equals the intake minus free energy in faeces and in all losses in relation to digestion. Unlike in typical static budgets, see {416}, the energy in urine is included in assimilation energy, because urine does not directly derive from food and is excreted by the organism, see {84} and {147}. (Faeces are not excreted, because they have never been inside the organism.)

The assimilation efficiency of food is here taken to be independent of the feeding rate. This makes the assimilation rate proportional to the ingestion rate, which seems to be realistic, see Figure 7.17. I later discuss the consistency of this simple assumption with more detailed models for enzymatic digestion, {266}. The conversion efficiency of food into assimilated energy is denoted by κ_X, so that $\{\dot{p}_{Am}\} = \kappa_X \{\dot{p}_{Xm}\}$, where $\{\dot{p}_{Am}\}$ stands for the maximum surface-area-specific assimilation rate. Both κ_X and $\{\dot{p}_{Am}\}$ are diet-specific parameters. The assimilated energy that comes in at food density X is now given by

$$\dot{p}_A = \kappa_X \dot{p}_X = \{\dot{p}_{Am}\} f L^2 \quad \text{where } f = \frac{X}{K + X} \tag{2.3}$$

and L the volumetric length. It does not involve the parameter $\{\dot{p}_{Xm}\}$ in the notation, which turns out to be useful in the discussion of processes of energy allocation in the next few sections.

2.3 Reserve dynamics

The change of the reserve energy E in time t can be written as $\frac{d}{dt}E = \dot{p}_A - \dot{p}_C$, where the assumption is that the mobilisation rate of reserve, $\dot{p}_C$, is some function of the amount of reserve energy E and of structural volume V only. This function is fully determined by the assumptions of strong and weak homeostasis, see {10}, but its derivation is not the easiest part of this book. Since mobilised reserve fuels metabolism, reserve dynamics is discussed in some detail. For the sake of parameter estimation from data that has no energy in its units, I will use the scaled reserve $U_E = E/\{\dot{p}_{Am}\}$, which has the un-intuitive dimension time $\times$ length^2, and treat it as 'something that is proportional to reserve energy'.

The dynamics for the reserve density has to be set up first, in general form. It can be written as the difference between the volume-specific assimilation rate, $[\dot{p}_A] = \dot{p}_A/V = f\{\dot{p}_{Am}\}/L$, and some function of the state variables: the reserve density $[E] = E/V$ and the structural volume V. The freedom of choice for this function is greatly restricted by the requirement that $[E]$ at steady state does not depend on size, while $[\dot{p}_A] \propto L^{-1}$. It implies that the dynamics can be written as

$$\frac{d}{dt}[E] = [\dot{p}_A] - L^{-1}\dot{H}([E]|\boldsymbol{\theta}) + ([E]^* - [E])\dot{G}([E], L) \tag{2.4}$$

where $\dot{H}([E]|\boldsymbol{\theta})$ is some function of $[E]$ and a set of parameters $\boldsymbol{\theta}$, that does not depend on L, and $\dot{G}([E], L)$ some function of $[E]$ and L. The value $[E]^*$ represents the steady-state reserve density, which can be found from $\frac{d}{dt}[E] = 0$. Since $[E]^*$ depends on food density

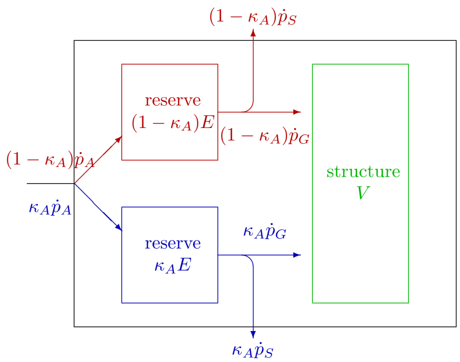


Figure 2.7: When the reserve is partitioned in two parts, the somatic maintenance costs and the costs for structure also need to be partitioned to arrive at a situation where the partitioning remains without effects for the individual.

via the assimilation power $[\dot{p}_A]$, the requirement that the rate of use of reserve should not depend on food density implies that $\dot{G}([E], L) = 0$ and (2.4) reduces to

$$\frac{d}{dt}[E] = [\dot{p}_A] - \dot{H}([E]|\boldsymbol{\theta})/L \tag{2.5}$$

The mass balance for the reserve density can be written as

$$\frac{d}{dt}[E] = [\dot{p}_A] - [\dot{p}_C] - [E]\dot{r} \tag{2.6}$$

where $\dot{r} = \frac{d}{dt} \ln V$ stands for the specific growth rate; the third term stands for the dilution by growth, which directly follows from the chain rule for differentiation of E/V. Because maintenance (work) and growth are among the destinations of mobilised reserve, the volume-specific mobilisation rate $[\dot{p}_C] = \dot{p}_C/V$ relates to these fluxes as $\kappa([E], V)[\dot{p}_C] = [\dot{p}_S] + [\dot{p}_G]$, or $\dot{r} = [\dot{p}_G]/[E_G] = (\kappa([E], V)[\dot{p}_C] - [\dot{p}_S])/[E_G]$, where the specific somatic maintenance costs $[\dot{p}_S]$ is some function of V and the volume-specific costs of structure $[E_G]$ is constant, in keeping with the homeostasis assumption for structural mass. The allocation fraction $\kappa([E], V)$ is, at this stage in the reasoning, some function of the state variables $[E]$ and V. Substitution of the expression for growth into (2.6) results in

$$\frac{d}{dt}[E] = [\dot{p}_A] - [\dot{p}_C](1 + \kappa[E]/[E_G]) + [E][\dot{p}_S](V)/[E_G] \tag{2.7}$$

Substitution of (2.5) leads to the volume-specific mobilisation rate

$$[\dot{p}_C] = \frac{\dot{H}([E]|\boldsymbol{\theta}^\circ)/L + [E][\dot{p}_S](V)/[E_G]}{1 + \kappa([E], V|\boldsymbol{\theta}^\circ)[E]/[E_G]} \tag{2.8}$$

where $\boldsymbol{\theta}^\circ$ is a subset of $\boldsymbol{\theta} = (\boldsymbol{\theta}^\circ, [\dot{p}_S], [E_G])$. Note that the functions $\dot{H}$ and κ cannot depend on $[\dot{p}_S]$ or $[E_G]$ because allocation occurs after mobilisation.

2.3.1 Partitionability follows from weak homeostasis

The next step in the derivation of reserve dynamics follows from the partitionability of reserve kinetics, meaning that the partitioning of reserve should not affect its dynamics, i.e. the sum of the dynamics of the partitioned reserves should be identical to that of the lumped one in terms of growth, maintenance, development and reproduction. Partitionability is implied by weak (and strong) homeostasis, as shown in [1085]. This originates from the fact that the reserves are generalised compounds, i.e. mixtures of various kinds of proteins, lipids, etc. Each of these compounds follows its own kinetics, which are functions of the amounts of that compound and of structural mass, but their relative amounts do not change; all reserve compounds have identical kinetics. The strong homeostasis assumption ensures that the amount of any particular compound of the reserve is a fixed fraction, say κ_A, of the total amount of reserve. This compound must account for a fraction κ_A of the maintenance costs and growth investment, see Figure 2.7.

In quantitative terms, partitionability means

$$\kappa_A[\dot{p}_C]([E], V|[\dot{p}_S], [E_G], \boldsymbol{\theta}^\circ) = [\dot{p}_C](\kappa_A[E], V|\kappa_A[\dot{p}_S], \kappa_A[E_G], \boldsymbol{\theta}^\circ) \tag{2.9}$$

for an arbitrary fraction κ_A in the interval $(0, 1)$. This factor not only applies to $[E]$, but also to the specific maintenance $[\dot{p}_S]$ and structure costs $[E_G]$, because the different fractions of the reserve contribute to these costs. The factor does not apply to V and the parameters $\boldsymbol{\theta}^\circ$. We can check in (2.8) that $[\dot{p}_C]$ is partitionable if

- the function $\dot{H}$ is a first-degree homogeneous function, which means that $\kappa_A\dot{H}([E]|\boldsymbol{\theta}^\circ) = \dot{H}(\kappa_A[E]|\boldsymbol{\theta}^\circ)$. It follows that this function can be written as $\dot{H}([E]) = \dot{v}[E]$, for some constant $\dot{v}$, which will be called the *energy conductance* (dimension length per time). The inverse, $\dot{v}^{-1}$, has the interpretation of a resistance. Conductances are often used in applied physics. Therefore, it is remarkable that the biological use of conductance measures seems to be restricted to plant physiology [561, 841] and neurobiology [673].

- the function κ is a zero-th degree homogeneous function in E, which means that $\kappa(\kappa_A[E], V) = \kappa([E], V)$. In other words: κ may depend on V, but not on $[E]$. Later, I argue that $\kappa(V)$ is a rather rudimentary function of V, namely a constant, see {40}.

Substitution of the function $\dot{H}$ into (2.5) gives

$$\frac{d}{dt}[E] = [\dot{p}_A] - [E]\dot{v}/L = (\{\dot{p}_{Am}\}f - [E]\dot{v})/L \tag{2.10}$$

The reserve density at steady state is $[E]^* = L[\dot{p}_A]/\dot{v} = f\{\dot{p}_{Am}\}/\dot{v}$. The maximum reserve density at steady state occurs at $f = 1$, which gives the relationship $[E_m] = \{\dot{p}_{Am}\}/\dot{v}$. So the reserve capacity $[E_m]$ represents the ratio of the assimilation and mobilisation fluxes. The scaled reserve density $e = [E]/[E_m]$ is a dimensionless quantity and has the simple dynamics

$$\frac{d}{dt}e = (f - e)\dot{v}/L \tag{2.11}$$

Notice that, if f is constant, e converges to f.

The reserve dynamics (2.10) results in the specific mobilisation and growth rates

$$[\dot{p}_C] = [\dot{p}_A] - \frac{d}{dt}[E] - [E]\dot{r} = [E](\dot{v}/L - \dot{r}) = [E]\frac{[E_G]\dot{v}/L + [\dot{p}_S]}{\kappa[E] + [E_G]} \tag{2.12}$$

$$\dot{r} = \frac{[E]\dot{v}/L - [\dot{p}_S]/\kappa}{[E] + [E_G]/\kappa} \tag{2.13}$$

After further specification of κ and $[\dot{p}_S]$, the mobilisation and specific growth rates are fully specified.

The mean time compounds stay in the reserve amounts to $t_E = E/\dot{p}_C$, which increases with length. For metabolically active compounds that lose their activity spontaneously, this phenomenon might contribute to the ageing process, to be discussed at {209}. This storage residence time must be large with respect to that of the stomach and the gut to justify neglecting the smoothing effect of the digestive tract.

If the reserve capacity, $[E_m]$, is extremely small, the dynamics of the reserve degenerates to $[E] = f[E_m]$, while both $[E]$ and $[E_m]$ tend to 0. The mobilisation rate then becomes $\dot{p}_C = \{\dot{p}_{Am}\}fV^{2/3}$. This case has been studied by Metz and Diekmann [785], but some consistency problems arise in variable environments where maintenance costs cannot always be paid.

2.3.2 Mergeability is almost equivalent to partitionability

Mergeability means that reserves can be added without effects on the reserve (density) dynamics if assimilation of the resources to synthesise the reserves is coupled and the total intake is constant. This is also implied by weak homeostasis; partitionable dynamics is also mergeable. This property is essential to understand the gradual reduction of the number of reserves during evolution, see {384}. The mergeability is also essential to understand symbiogenesis in a DEB theory context, see {391}: Given that species 1 and 2 each follow DEB rules, and species 1 evolves into an endosymbiont of species 2, the new symbiosis again should follow the DEB rules [647] (otherwise the theory becomes species-specific), see {391}. Evolution might have found several mechanisms to obtain mergeability of reserves, but the fact that they are mergeable is essential for evolution, see {380}.

A quantitative definition of mergeability is as follows. Given $\frac{d}{dt}[E_i] = [\dot{p}_{A_i}] - \dot{F}([E_i], V)$ for $i = 1, 2, \cdots$ and $[\dot{p}_{A_i}] = \kappa_{A_i}[\dot{p}_A]$ with $\sum_i \kappa_{A_i} = 1$, two reserves E_1 and E_2 are mergeable if $\frac{d}{dt}\sum_i[E_i] = [\dot{p}_A] - \dot{F}(\sum_i[E_i], V)$. The mergeability condition summarises to $\sum_i \dot{F}([E_i], V) = \dot{F}(\sum_i[E_i], V)$.

Weak homeostasis implies that $\dot{F}([E], V) = V^{-1/3}\dot{H}([E])$, see (2.5), so together with the mergeability requirement this translates into the requirement that $\sum_i \dot{H}([E_i]) = \dot{H}(\sum_i[E_i])$ or $\kappa_A\dot{H}([E]) = \dot{H}(\kappa_A[E])$ for an arbitrary positive value of κ_A. In other words: H must be first–degree homogeneous in $[E]$. From this follows directly $\dot{H}([E]) = \dot{v}[E]$.

Since from partitionability also follows that κ is a zero-th degree homogeneous function in E, while this does not follow from mergeability, the latter requirement is less restrictive. In other words, partitionability imposes constraints on the fate of mobilised reserve, mergeability does not. More specifically, partitionability involves maintenance explicitly, mergeability does not.

To demonstrate the difference I now translate the mergeability constraint on $\dot{F}$ to a constraint on the mobilisation flux $\dot{p}_C$. These two fluxes relate to each other as $\dot{F} = [\dot{p}_C] + [E]\dot{r}$, where the specific growth rate $\dot{r} = [\dot{p}_G]/[E_G] = (\kappa[\dot{p}_C] - [\dot{p}_S])/[E_G]$. So

$$\dot{F} = [\dot{p}_C] + (\kappa[\dot{p}_C] - [\dot{p}_S])[E]/[E_G] = \left(1 + \frac{\kappa[E]}{[E_G]}\right)[\dot{p}_C] - \frac{[E]}{[E_G]}[\dot{p}_S] \qquad (2.14)$$

The mergeability constraint $\kappa_A \dot{F}([E], V) = \dot{F}(\kappa_A[E], V)$ can be written as

$$\kappa_A[\dot{p}_C]([E], V) = [\dot{p}_C](\kappa_A[E], V)\frac{[E_G] + \kappa_A\kappa[E]}{[E_G] + \kappa[E]} \tag{2.15}$$

which can now be compared with the partitionability definition (2.9).

2.3.3 Mechanism for mobilisation and weak homeostasis

Although Lemesle and Mailleret [695] correctly observed that the use of food and reserve are both Michaelis–Menten functions of their densities, and the mechanism behind the use of food is known, the use of reserve must have a different mechanism because the parameters have a specific interpretation and reserve and structure are not homogeneously mixed. Also first-order kinetics, which is very popular in chemistry, cannot apply because it is not partitionable for isomorphs. Finding a mechanism for reserve mobilisation has been a challenge; an elegant and realistic mechanism for the reserve dynamics rests on the structural homeostasis concept, see {10}. The arguments are as follows.

Since reserve primarily consists of polymers (RNA, proteins, carbohydrates) and lipids, an interface exists between reserve and structure and the gross mobilisation rate of reserve is now taken to be proportional to the surface area of the reserve–structure interface, so $E\dot{v}/L$, see Figure 1.4, and allocated to the mobilisation Synthesising Units (SUs). This flux is partitioned for $\dot{r} \geq 0$ into a net mobilisation flux $\dot{p}_C = E\dot{v}/L - E\dot{r}$ that is accepted by the mobilisation SUs and used for metabolism, and a flux $E\dot{r}$ is rejected and fed back to the reserve. This particular partitioning follows from the weak homeostasis argument, where the ratio of formation of reserve, $\dot{r}E$, and structure, $\dot{r}V$, equals the existing ratio of both amounts, E/V, and the rejected flux is formally considered as a 'synthesis' of reserve. From (2.6) now directly follows (2.10).

A beautiful property of this mechanism is that the correct partitioning of the gross mobilisation flux automatically follows from SUs kinetics, see {98}, if the specific number of mobilisation SUs (in C-moles) equals $\frac{y_{EV}\dot{v}}{L\dot{k}}$, where y_{EV} is the yield of reserve on structure (in C-moles), and $\dot{k}$ is the (constant) dissociation rate of the SUs [661]. An increased deviation from this value results in an increased deviation from weak homeostasis, and the selection of the proper value possibly links directly to the evolution of weak homeostatis. If membranes wrap reserve vesicles, the SU density in these membranes would be constant. Strong homeostasis can only apply strictly if mobilisation SUs switch to the active state if bound to these membranes.

This mechanism has a problem for embryos (eggs, seeds), because L is initially very small, so the gross mobilisation rate as well as the rejected flux are very large. This is unrealistic, because the absence of respiration in the early embryo excludes substantial metabolic activity. This problem hardly exists for organisms that propagate by division, so it became a problem when embryos evolved during evolution. Also organisms with a large body size suffer from the problem, because they have a relatively large amount of reserve, see {292}. This problem can be solved by a self-inhibition mechanism for monomerisation; polymers as such do not take part in metabolism as substrates. The SUs take their substrate from a small pool of reserve monomers, and the pool size of the reserve monomers is proportional to that of the reserve polymers; this is implied by the strong homeostasis assumption.

The self-inhibition mechanism might be by rapid interconversion of the first–order type, but this comes with metabolic costs. A more likely possibility is that monomerisation is product inhibited and ceases if the monomers per polymer reach a threshold. The monomerisation cost is then covered by maintenance and growth. For an individual with an amount of structure M_V and reserve M_E, the kinetics of the amount of monomers M_F could be

$$\frac{d}{dt}M_E = -M_E(\dot{k}_{EF} - \frac{m_F}{m_E}\dot{k}_{FE}); \quad \frac{d}{dt}M_F = y_{FE}M_E(\dot{k}_{EF} - \frac{m_F}{m_E}\dot{k}_{FE}) \tag{2.16}$$

with $m_E = M_E/M_V$ and $m_F = M_F/M_V$. This kinetics makes that in steady state $\frac{m_F^*}{m_E^*} = \frac{\dot{k}_{EF}}{\dot{k}_{FE}}$. The monomerisation occurs at the E–V interface, which has a surface area proportional to E/L in isomorphs. This means that k_{EF} and k_{FE} are proportional to L^{-1} as well.

2.4 The κ-rule for allocation to soma

The motivation for a κ-rule originates from the maturation concept, see {44}, which implies four destinies of mobilised reserve: growth plus somatic maintenance, summarised by the term *soma*, and maturation (or reproduction) plus maturity maintenance. The partitionability requirement states that the fraction κ of mobilised reserve that is allocated to the soma cannot depend on the amount of reserve (or on the reserve density) but still can depend on the amount of structure. The simplest assumption is that κ is constant and does also not depend on the amount of structure. This assumption is used by the standard DEB model.

The empirical evidence for a constant value of κ is that then the von Bertalanffy growth curve results at constant food density, see {49}, which typically fits data from many species very well. Even stronger support is provided by the resulting body size scaling relationships, which are discussed in Chapter 8. Moreover Huxley's allometric model for relative growth of body parts closely links up with multivariate extensions of this κ-rule, see {196}. Strong support for the κ-rule also comes from situations where the value for κ is changed to a new fixed value. Such a simple change affects reproduction as well as growth and so food intake in a very special way. Parasites such as the trematode *Schistosoma* in snails harvest all energy to reproduction and increase κ to maximise the energy flow they can consume; see [563] for a detailed physiological discussion. Parasite-induced gigantism, coupled to a reduction of the reproductive output, is also known from trematode-infested chaetognaths (*Sagitta*) [820], for instance. The daily light cycle also affects the value for κ in snails, and the allocation behaviour during prolonged starvation; see {113}. The effect of some toxic compounds can be understood as an effect on κ, as is discussed on {235}.

A possible mechanism for a constant κ is as follows. At separated sites along the path the blood follows, somatic cells and ovary cells pick up energy. The only information the cells have is the energy content of the blood and body size, see {12}. They do not have information about each other's activities in a direct way. This also holds for the mechanism by which energy is added to or taken from energy reserve. The organism only has information on the energy density of the blood, and on size, but not on which cells have removed energy from the blood. This is why the parameter κ does not show up in the dynamics of reserve density. The activity of all carriers that remove energy from the body fluid and transport it across the cell membrane depends, in the same way, on the energy density of the fluid. Somatic cells and ovary cells both may use the same carriers, but the concentration in their membranes may differ so that $1-\kappa$ may differ from the ratio of ovary to body weight. This concentration of active carriers is controlled, by hormones for example, and depends on age, size and environment. Once in a somatic cell, energy is first used for maintenance, the rest is used for growth. This makes maintenance and growth compete directly, while development and reproduction compete with growth plus maintenance at a higher level. The κ-rule makes growth and development parallel processes that interfere only indirectly, as is discussed by Bernardo [89], for instance.

If conditions are poor, the system can block allocation to reproduction, while somatic maintenance and growth continue to compete in the same way, see {113}.

The κ-rule solves quite a few problems from which other allocation rules suffer. Although it is generally true that reproduction is maximal when growth ceases, a simple allocation shift from growth to reproduction leaves similarity of growth between different sexes unexplained, since the reproductive effort of males is usually much less than that of females. The κ-rule implies that size control is the same for males and females and for organisms such as yeasts and ciliates, which do not spend energy on reproduction but do grow in a way that is comparable to species that reproduce; see Figure 2.14. It is important to realise that although the fraction of mobilised reserve that is allocated to the soma remains constant, the absolute size of the flow tends to increase during growth at constant food densities.

The value of κ can be extracted from combined observations on growth and reproduction. The observed value of 0.8 for *D. magna* in Figure 2.10 is well above the value that would maximise the reproduction rate, given the other parameter values. This questions the validity of maximisation of fitness without much attention for mechanisms, a line of thinking that has become popular among evolutionary biologists, e.g. [980, 1099]. The reason why the measured value of κ is so high is an open problem, but might be linked with the length at birth and at puberty, which increases sharply with κ in the standard DEB model.

2.5 Dissipation excludes overheads of assimilation and growth

Dissipation is defined as the use of reserve that is not coupled to net production, where reproduction is seen as an excretion process, rather than a production process in the strict sense of the word. Assimilation and growth have overheads that also appear in the environment, and are excluded from dissipation. So dissipation is not all that dissipates,

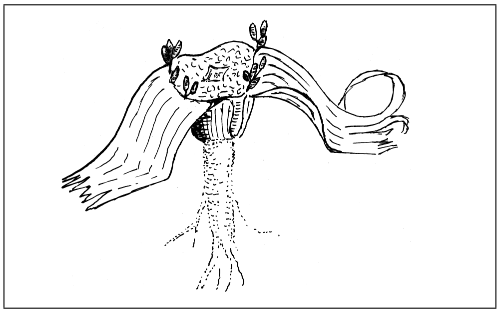


Figure 2.8: The leaves of most plant species grow during a relatively short time period, and are shed yearly, after the plant has recovered useful compounds. The leaves of some species, however, such as *Welwitschia mirabilis*, grow and weather continuously. The life span of this remarkable gymnosperm can exceed 2000 years. Leaves have a very limited functional life span, but plants have found different ways to deal with that problem.

but less than that. The reason to group these fluxes together under one label, dissipation, becomes clear in the discussion on the organisation of mass fluxes.

Dissipation represents metabolic work. Reserve is metabolised and the metabolites are generally excreted into the environment, frequently in mineralised form (mainly water, carbon dioxide and ammonia), see {159}. Dissipation has four components: somatic and maturity maintenance, maturation and reproduction overhead.

2.5.1 Somatic maintenance is linked to volume and surface area

Maintenance costs can generally be decomposed in contributions that are proportional to structural body volume, and to surface area, which gives the quantification

$$[\dot{p}_S] = [\dot{p}_M] + \{\dot{p}_T\}/L \qquad (2.17)$$

where the costs that comprise the volume-linked maintenance costs $[\dot{p}_M]$ and the surface-area-linked maintenance costs $\{\dot{p}_T\}$ are discussed below.

Volume-linked maintenance costs

Maintenance processes include the maintenance of concentration gradients across membranes, the turnover of structural body proteins, a certain (mean) level of muscle tension and movement, and the (continuous) production of hairs, feathers, scales, leaves (of trees), see Figure 2.8.

The idea that maintenance costs are proportional to biovolume is simple and rests on strong homeostasis: a metazoan of twice the volume of a conspecific has twice as many cells, which each use a fixed amount of energy for maintenance. A unicellular of twice the original volume has twice as many proteins to turn over. Protein turnover seems to be low in prokaryotes [612]. Another major contribution to maintenance costs relates to the maintenance of concentration gradients across membranes. Eukaryotic cells are filled with membranes, and this ties the energy costs for concentration gradients to volume. (The argument for membrane-bound food uptake works out differently in isomorphs, because feeding involves only the outer membrane directly.) Working with mammals, Porter and Brand [912] argued that proton leak in mitochondria represents 25 % of the basal respiration in isolated hepatocytes and may contribute significantly to the standard metabolic rate of the whole animal.

The energy costs of movement are also taken to be proportional to volume if averaged over a sufficiently long period. Costs of muscle tension in isomorphs are likely to be proportional to volume, because they involve a certain energy investment per unit volume of muscle. In the section on feeding, I discuss briefly the energy involved in movement, {31}, which has a standard level that includes feeding. This can safely be assumed to be a small fraction of the total maintenance costs. Sustained powered movement such as in migration requires special treatment. Such activities involve temporarily enhanced metabolism and feeding. The occasional burst of powered movement hardly contributes to the general level of maintenance energy requirements. Sustained voluntary powered movement seems to be restricted to humans and even this seems of little help in getting rid of weight!

There are many examples of species-specific maintenance costs. Daphnids produce moults every other day at 20 °C. The synthesis of new moults occurs in the intermoult period and is a continuous and slow process. The moults tend to be thicker in the larger sizes. The exact costs are difficult to pin down, because some of the weight refers to inorganic compounds, which might be free of energy cost. Larvaceans produce new feeding houses every 2 hours at 23 °C [340], and this contributes substantially to organic matter fluxes in oceans [17, 18, 248]. These costs are taken to be proportional to volume. The inclusion of costs of moults and houses in maintenance costs is motivated by the observation that these rates do not depend on feeding rate [340, 633], but only on temperature.

It will be convenient to introduce the maintenance rate coefficient $\dot{k}_M = [\dot{p}_M]/[E_G]$ as compound parameter. It stands for the ratio of the costs of somatic maintenance and of structure and has dimension time^{-1}. It was introduced by Marr *et al.* [745] for the first time and publicised by Pirt [899].

Surface-area-linked maintenance costs

Some specialised maintenance costs relate to surface areas of individuals.

Aquatic insects are chemically fairly well isolated from the environment. Euryhaline fishes, however, have to invest energy in osmoregulation when in waters that are not iso-osmotic. The cichlid *Oreochromis niloticus* is iso-osmotic at 11.6 ‰ and 29 % of the respiration rate at 30 ‰ can be linked to osmoregulation [1271]. Similar results have been obtained for the brook trout *Salvelinus fontinalis* [376].

Endotherms (birds, mammals) use reserve to heat their body such that a particular part of the body (the head and the heart region in humans) has a constant temperature during the post-embryonic stages; embryos don't do this. Heat loss is not only proportional to surface area but, according to I. Newton, also to the temperature difference between body and environment. This is incorporated in the concept of thermal conductance $\{\dot{p}_T\}/(T_e - T_b)$, where T_e and T_b denote the temperature of the environment and the body. It is about 5.43 J cm^{-2} h^{-1} °C^{-1} in birds and 7.4–9.86 J cm^{-2} h^{-1} °C^{-1} in mammals, as calculated from [496]. The unit cm^{-2} refers to volumetric squared length, not to real surface area which involves shape. The values represent crude means in still air. The thermal conductance is roughly proportional to the square root of wind speed.

This is a simplified presentation. Birds and mammals moult at least twice a year, to replace their hair and feathers which suffer from wear, and change the thick winter coat for the thin summer one. Cat owners can easily observe that when their pet is sitting in the warm sun, it will pull its hair into tufts, especially behind the ears, to facilitate heat loss. Many species have control over blood flow through extremities to regulate temperature. People living in temperate regions are familiar with the change in the shape of birds in winter to almost perfect spheres. This increases insulation and generates heat from the associated tension of the feather muscles. These phenomena point to the variability of thermal conductance.

There are also other sources of heat exchange, through ingoing and outgoing radiation and cooling through evaporation. Radiation can be modulated by changes in colour, which chameleons and tree frogs apply to regulate body temperature [718]. Evaporation obviously depends on humidity and temperature. For animals that do not sweat, evaporation is tied to respiration and occurs via the lungs. Most non-sweaters pant when hot and lose heat by enhanced evaporation from the mouth cavity. A detailed discussion of heat balances would involve a considerable number of coefficients [802, 1090], and would obscure the main line of reasoning. I discuss heating in connection with the water and energy balances on {155, 154}. It is important to realise that all these processes are proportional to surface area, and so affect the heating rate $\{\dot{p}_T\}$ and in particular its relationship with the temperature difference between body and environment.

It turns out to be convenient to introduce the heating length $L_T \equiv \{\dot{p}_T\}/[\dot{p}_M]$ or the heating volume $V_T \equiv L_T^3$ as compound parameters of interest. Heating length stands for the reduction in length endotherms experienced due to the energy costs of heating. It can be treated as a simple parameter as long as the environmental temperature remains constant. Sometimes, it will prove to be convenient to work with the scaled heating length $l_T \equiv \{\dot{p}_T\}/\kappa\{\dot{p}_{Am}\}$ as a compound parameter. If the temperature changes slowly relative to the growth rate, the heating volume is a function of time. If environmental temperature changes rapidly, body temperature can be taken to be constant again while the effect contributes to the stochastic nature of the growth process, see {111}.

2.5.2 Maturation for embryos and juveniles

The κ-rule makes that growth and development are parallel processes, which links up beautifully with the concepts of acceleration and retardation of developmental phenomena such as sexual maturity [422]. These concepts are used to describe relative rates of development in species that are similar in other respects.

The ideas on maturation and maturation maintenance in the DEB context rest on four observations:

- Contrary to age, the volume of the mother at the first appearance of eggs hardly depends on food density, typically; see Figure 2.9. The same holds for the volume at birth.

- Some species, such as daphnids, continue to grow after the onset of reproduction. *Daphnia magna* starts to reproduce at a length of 2.5 mm, while its ultimate size is 5 mm, if well fed. This means an increase of well over a factor eight in volume during

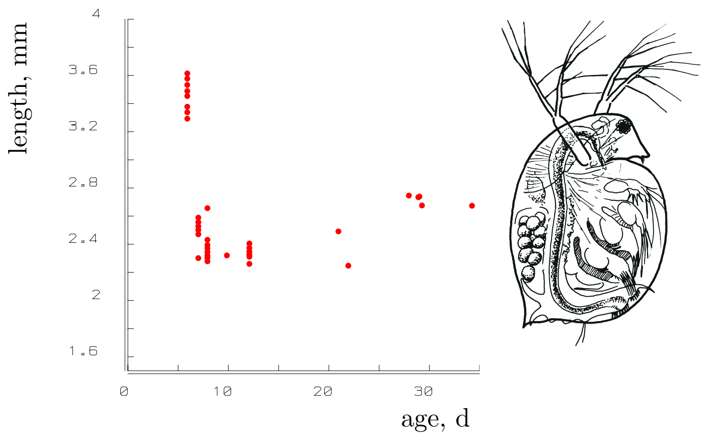


Figure 2.9: The carapace length of the daphnid *Daphnia magna* at 20 °C for 5 different food levels at the moment of egg deposition in the brood pouch. Data from Baltus [57]. The data points for short juvenile periods correspond with high food density and growth rate. They are difficult to interpret because length increase is only possible at moulting in daphnids.

the reproductive period. Other species, however, such as birds, only reproduce well after the growth period. The giant petrel wanders 7 years over Antarctic waters before it starts to breed for the first time. This means that stage transitions cannot be linked to size.

- The total cumulative energy investment in development at any given size of the individual depends on food density. Indeed, if feeding conditions are so poor that the ultimate volume is less than the threshold for allocation to reproduction, the cumulated energy investment in development becomes infinitely large if survival allows.

- If food density is constant, the reproduction rate at ultimate size is a continuous function of the food density; it is zero for low food densities, and increases from zero for increasing food densities. So it does not make a big jump if the various food densities differ sufficiently little.

The combination of the four graphs in Figure 2.10 illustrates a basic problem for the rules for allocating energy quantitatively. The problem becomes visible as soon as one realises that a considerable amount of energy is invested in reproductive output. The volume of young produced exceeds one-quarter of that of the mother each day. The problem is that growth is not retarded in animals crossing the 2.5 mm barrier; they do not feed much more and simply follow the surface area rule with a fixed proportionality constant at constant food densities; they do not change sharply in respiration, so it seems unlikely that they digest their food much more efficiently. So where does the energy allocated to reproduction come from?

These observations fit naturally if stage transitions are linked to maturity and a maturity maintenance flux exists that is proportional to the level of maturity. The recognition of the problem and its solution are cornerstones of DEB theory.

A solution to this problem can be found in maturation. Juvenile animals have to mature and become more complex. They have to develop new organs and install regulation systems. The increase in size (somatic growth) of the adult does not include an increase in complexity. The energy spent on development in juveniles is spent on reproduction in

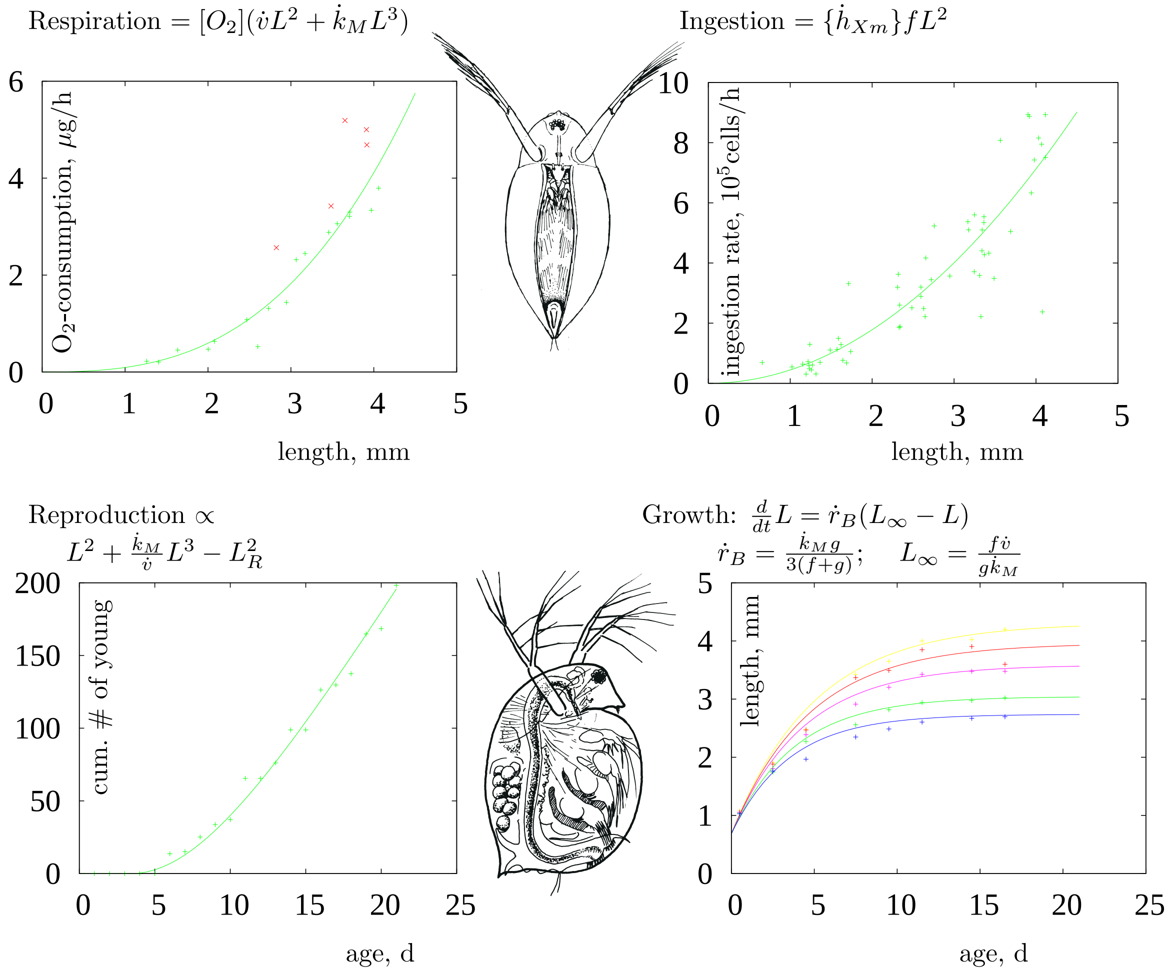


Figure 2.10: Respiration (upper left), and ingestion (upper right) as a function of body length, and reproduction (lower left) at high food level and body length (lower right) as function of age at different food levels in the waterflea *Daphnia magna* at 20 °C. Original data and from [330, 651]; the DEB model specifies the curves. The reproduction curve shows that *D. magna* starts to reproduce at the age of 7 d, i.e. when its length exceeds 2.5 mm. However, respiration, and ingestion do not increase steeply at this size, nor does growth decrease. Where did the substantial reproductive energy come from? The κ-rule gives the explanation. The open symbols in the graph for respiration relate to individuals with eggs in their brood pouch. Parameter values: $\delta_{\mathcal{M}} = 0.54$ (fixed), $\kappa = 0.799$, $\kappa_R = 0.95$ (fixed), $g = 0.15$, $\dot{k}_J = 3.57\,\mathrm{d}^{-1}$, $\dot{k}_M = 4.06\,\mathrm{d}^{-1}$, $\dot{v} = 1.62\,\mathrm{mm\,d}^{-1}$, $U_H^b = 0.001\,\mathrm{mm^2d}$, $U_H^p = 0.049\,\mathrm{mm^2d}$, $[O_2] = 2.033\,\mu\mathrm{g\,mm}^{-3}$, $\{\dot{h}_{Xm}\} = 1.53\,10^5\mathrm{cells\,h^{-1}mm^{-2}}$, $f = 0.88, 0.81, 0.73, 0.63, 0.56$ in lower-right graph. The observation that $L_b = 0.8$ mm and $L_p = 2.5$ mm at $f = 1$ and 0.5 has been used to stabilise the estimate for $\dot{k}_J$. The calculated lengths are $L_b = 0.686, 0.685$ mm and $L_p = 2.46, 2.45$ mm, respectively. All length measures in the parameters are volumetic. The shape coefficient $\delta_{\mathcal{M}}$ affects physical lengths and is probably too large because of the water pockets inside daphnid's carapace. A multiplication of $\delta_{\mathcal{M}}$ by a factor z means a multiplication of $\dot{v}$ by z, of U_H^b and U_H^p by z^2, of $[O_2]$ by z^{-3}, and of $\{\dot{h}_{Xm}\}$ by z^{-2}. This can be seen from the units of the parameters.

adults. This switch does not affect growth and suggests the 'κ-rule': a fixed fraction κ of energy mobilised from the reserves is spent on somatic maintenance plus growth, the remaining fraction $1-\kappa$ on maturity maintenance plus maturation plus reproduction. The partitionability of reserve kinetics has led to the conclusion that κ cannot depend on the reserve density, see {37}. The argument that allocation is an intensive process, not an extensive one, suggests that κ is independent of V as well.

If the maturity maintenance rate coefficient equals the somatic one, the maturity density, so the ratio of the maturity and the amount of structure, remains constant and stage transitions then also occur at fixed amounts of structure, see {51}. Some species do show variations in the size at first reproduction, however, see [89]. Little is known about the molecular machinery that is involved in the transition from the juvenile to the adult stage. Recent evidence points to a trigger role of the hormone leptin in mice, which is excreted by the adipose tissue [203]. This finding supports the direct link between the transition and energetics.

The increase in the level of maturity, quantified as cumulative investment of reserve into maturity equals

$$\frac{d}{dt}E_H = \dot{p}_R(E_H < E_H^p) \quad \text{with } \dot{p}_R = (1-\kappa)\dot{p}_C - \dot{p}_J \tag{2.18}$$

where E_H^p is the maturity threshold at puberty. Because of the arbitrariness of a unit for information, I refrain from an explicit conversion to information, but the mass as well as the energy in this investment is typically excreted into the environment in the form of metabolites and heat.

The literature distinguishes determinate growers, which cease growth during the adult stage, and indeterminate growers, which continue growth. In DEB theory, this is just a matter of the value of E_H^p relative to other DEB parameters; even for ecdysozoans that reproduce only in their final moult, typically follow von Bertalanffy growth curves, see {49}. The only real determinate growers in the DEB context are the holometabolic insects, which insert a pupal stage between the juvenile and adult stages, see {277}, and don't grow as adults; their growth as larvae to pupation is not asymptotic.

2.5.3 Maturity maintenance: defence systems

The maturity maintenance is assumed to be proportional to the maturity level

$$\dot{p}_J = \dot{k}_J E_H \tag{2.19}$$

where $\dot{k}_J$ is the maturity maintenance rate coefficient. It can be compared with the somatic maintenance rate coefficient $\dot{k}_M$, and we will see, {51}, that if the maintenance ratio $k = \dot{k}_J/\dot{k}_M = 1$ and the heating length $L_T = 0$, stage transitions also occur at fixed structural volumes. Notice that $\dot{k}_M$ expresses the maintenance costs relative to the cost for structure; likewise $\dot{k}_J$ expresses the maintenance costs relative to the cost of a unit of maturity, but since we quantify maturity as the cumulative reserve investment into maturity this unit equals 1 by definition; we don't make the conversion to maturity as information explicitly.

An observation that strongly supports the existence of maturity maintenance concerns pond snails, where the day/night cycle affects the fraction of utilised energy spent on maintenance plus growth [1292] such that κ at equal day/equal night, κ_{md}, is larger than that at long day/short night, κ_{ld}. Apart from the apparent effects on growth and reproduction rates, volume at the transition to adulthood is also affected. If the cumulated energy investment in the increase of maturity does not depend on the value for κ and if the maturity maintenance costs are $\dot{p}_J = \frac{1-\kappa}{\kappa}\dot{p}_M$, the expected effect is $\frac{V_{p,ld}}{V_{p,md}} = \frac{\kappa_{ld}(1-\kappa_{md})}{\kappa_{md}(1-\kappa_{ld})}$, which is consistent with the observations on the coupling of growth and reproduction investments to size at puberty [1292].

Maturity maintenance can be thought to relate to the maintenance of regulating mechanisms and concentration gradients, such as those found in *Hydra*, that maintain head/foot differentiation [405].

During extreme forms of starvation, many organisms shrink {118}. They can only recover enough energy from the degradation of structural mass to pay the somatic maintenance costs if they can reduce the maturity maintenance costs under those conditions. Thomas and Ikeda [1148] concluded from studies on laboratory populations of *Euphausia superba* that female krill can regress from the adult to the juvenile state during starvation.

Organisms also become more vulnerable to diseases during starvation. This suggests that defence systems, see {382}, such as the immune system of vertebrates are fuelled from maturity maintenance, and that maturity maintenance is more facultative than somatic maintenance. All species have several defence systems, also to protect themsevels against effects of toxicants. This also explains why maturity maintance can be substantial.

2.5.4 Reproduction overhead

Since embryos initially consist almost exclusively of reserve, the allocation to reproduction consists of reserve, and the strong homeostasis assumption means that reserve cannot change in composition, little metabolic work is involved in reproduction. Yet some work is involved in the conversion of (part of) the buffer of reserve that is allocated to reproduction into eggs. A fraction κ_R of the reproduction flux, called the reproduction efficiency, is assumed to be fixed in embryo reserve, and the rest, a fraction $1-\kappa_R$, is used as reproduction overhead.

2.6 Growth: increase of structure

Now that the allocation fraction κ and the specific maintenance costs $[\dot{p}_S]$ are specified, the specific mobilisation (2.12) and growth rates (2.13) can be written as

$$[\dot{p}_C] = [E_m](\dot{v}/L + \dot{k}_M(1 + L_T/L))\frac{eg}{e+g} \tag{2.20}$$

$$\frac{d}{dt}V = \dot{r}V \quad \text{with } \dot{r} = \dot{v}\frac{e/L - (1 + L_T/L)/L_m}{e+g} \tag{2.21}$$

where the energy investment ratio $g = [E_G]/\kappa[E_m]$ stands for the costs of new biovolume relative to the maximum potentially available energy for growth plus maintenance. It is dimensionless.

The maximum length $L_m = \kappa\{\dot{p}_{Am}\}/[\dot{p}_M] = \dot{v}/\dot{k}_M g$ represents the ratio of the part of the assimilation flux that is allocated to the soma and the somatic maintenance flux. Notice the conceptual similarity with the reserve capacity $[E_m] = \{\dot{p}_{Am}\}/\dot{v}$, which is also a ratio of in- and out-going fluxes. The maximum volume $V_m = L_m^3$ is also a compound parameter of interest.

On Gough Island, the house mouse *Mus musculus* changed diet and turned to prey on the chicks of the Tristan albatross *Diomedea dabbenena* and the Atlantic petrel *Pterodroma incerta*, despite the fact that these birds are 250 times heavier. From an energy point of view, this had the remarkable effect that the weight of the adult mice is 40 g, rather than the typical 15 g. The reason is probably that the conversion efficiency from birds to mice is higher than their typical conversion efficiency. This supports the idea that ultimate (structural) weight represents the ratio of assimilation and maintenance. From a nature conservation point of view the problem is that 99 % of the world population of these two bird species live on this island; the birds are now threatened with extinction.

For some applications, it will be convenient to work in scaled length $l = L/L_m$ and with $\frac{d}{dt}l = l\dot{r}/3$ and scaled reserve density e. From (2.11) and (2.21) we have

$$\frac{d}{dt}l = \dot{k}_M \frac{g}{3} \frac{e - l - l_T}{e + g} \quad \text{and} \quad \frac{d}{dt}e = (f - e) g \dot{k}_M / l \tag{2.22}$$

Animals that have non-permanent exoskeletons, the Ecdysozoa, have to moult to grow. The rapid increase in size during the brief period between two moults relates to the uptake of water or air, not to synthesis of new structural biomass, which is a slow process occurring during the intermoult period. This minor deviation from the DEB model relates more to size measures than to model structure.

2.6.1 Von Bertalanffy growth at constant food

If food density X and, therefore, the scaled functional response f, are constant, and if the initial reserve density equals $[E] = f[E_m]$, reserve density will not change and $e = f$. Volumetric length as a function of time since birth can then be solved from $\frac{d}{dt}L = L\dot{r}/3$ and results in

$$L(t) = L_\infty - (L_\infty - L_b)\exp(-t\dot{r}_B) \quad \text{or } t(L) = \frac{1}{\dot{r}_B} \ln \frac{L_\infty - L_b}{L_\infty - L} \tag{2.23}$$

$$\dot{r}_B = \frac{1}{3/\dot{k}_M + 3fL_m/\dot{v}} = \frac{\dot{k}_M/3}{1 + f/g} \tag{2.24}$$

$$L_\infty = fL_m - L_T \tag{2.25}$$

where length at birth $L_b \equiv L(0)$ is a quantity that will be discussed at {60}. See Figure 2.14 for a graphical interpretation of the $L(t)$ curve. I will follow tradition and call this curve the von Bertalanffy growth curve despite its earlier origin and von Bertalanffy's contribution of introducing allometry, which I reject. Equations (2.24) and (2.25) give a

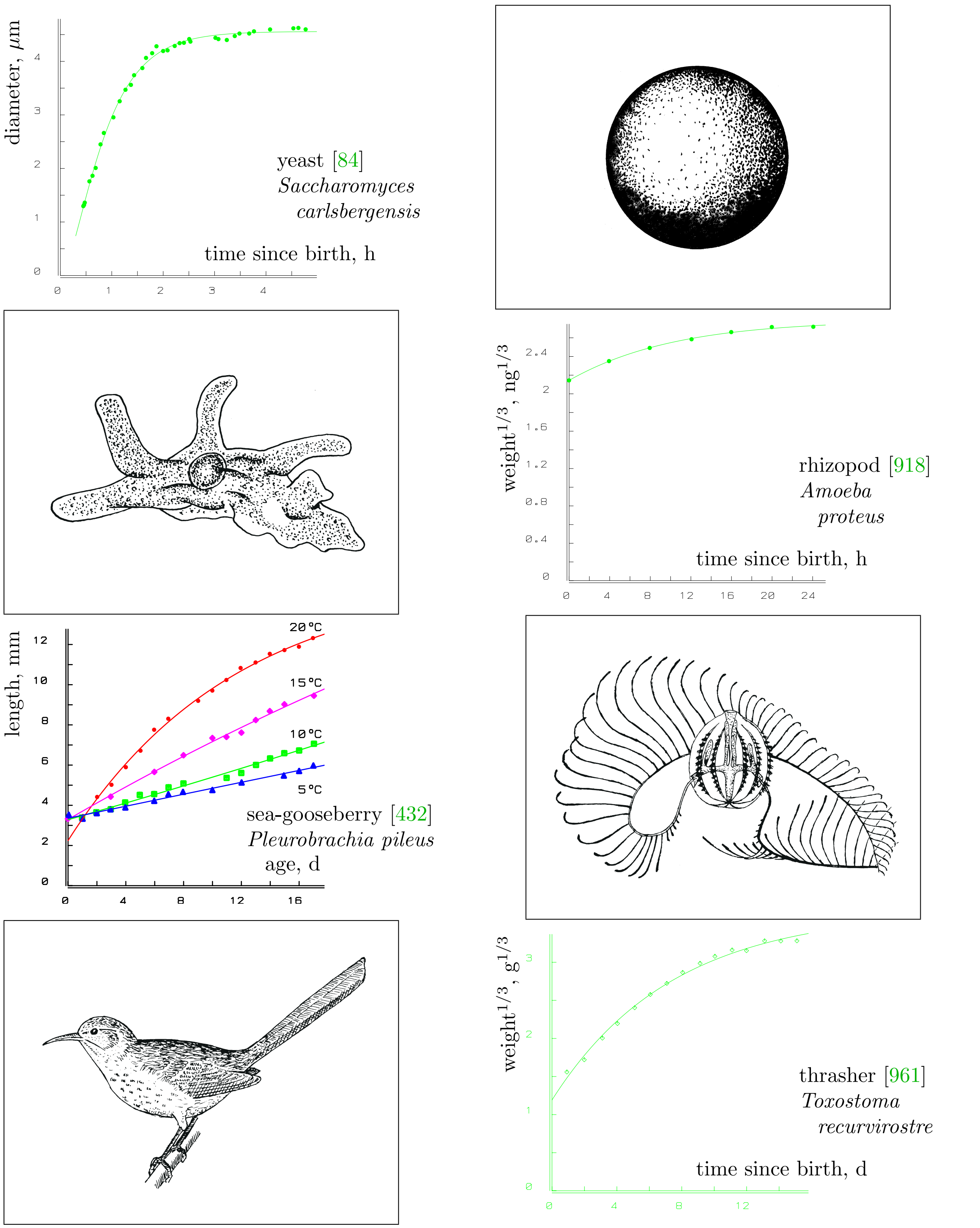


Figure 2.11: These von Bertalanffy growth curves fit very well data of organisms that differ considerably in their growth–regulating systems. This suggests that hormones are used to match local supply and demand of metabolites, but growth is controlled at the level of the individual, including hormonal activity.

physiological interpretation of the von Bertalanffy growth rate $\dot{r}_B$ and the ultimate length L_∞.

The von Bertalanffy growth curve results for post-natal isomorphs at constant food density and temperature and has been fitted successfully to the data of some 270 species from many different phyla, which have very different hormonal systems to control growth; see Figure 2.11 and Table 8.3. The gain in insight since A. Pütter's original formulation in 1920 [?] is in the interpretation of the parameters in terms of underlying processes. It appears that heating cost does not affect the von Bertalanffy growth rate $\dot{r}_B$. Food density affects both the von Bertalanffy growth rate and the ultimate length. The inverse of the von Bertalanffy growth rate is a linear function of the ultimate volumetric length; see Figure 2.10. This is in line with Pütter's original formulation, which took this rate to be inversely proportional to ultimate length, as has been proposed again by Gallucci and Quinn [384]. DEB theory shows, however, that the intercept cannot be zero.

The requirement that food density is constant for a von Bertalanffy curve can be relaxed if food is abundant, because of the hyperbolic functional response. As long as food density is higher than four times the saturation coefficient, food intake is higher than 80 % of the maximum possible food intake, which makes it hardly distinguishable from maximum food intake. Since most birds and mammals have a number of behavioural traits aimed at guaranteed adequate food availability, they appear to have a fixed volume–age relationship. This explains the popularity of age-based models for growth in 'demand' systems. Later, on {167}, I discuss deviations from the von Bertalanffy growth curve that can be understood in the context of the present theory.

In contrast, at low food densities, fluctuations in food density soon induce deviations from the von Bertalanffy curve. This phenomenon is discussed further in the section on genetics and parameter variation, {288}. Growth ceases, i.e. $\frac{d}{dt}V = 0$, if the reserve density equals a threshold value, $[E] = [\dot{p}_S]L/\kappa\dot{v}$.

2.6.2 States at birth and initial amount of reserve

During the embryo stage, dry weight decreases, but the amount of structure increases: reserve is converted into structure. The initial amount of reserve is not a free parameter because the reserve density, i.e. the ratio of the amounts of reserve and structure, at birth tends to covary with that of the mother at egg production; well-fed mothers give birth to well-fed offspring. Such maternal effects are typical and have been found in e.g. birds [823], reptiles, amphibians [714], fishes [477], insects [769, 992, 991], crustaceans [410], rotifers [1285], echinoderms and bivalves [98]. Maternal effects explain, for instance, why the batch fecundity of the anchovy *Engraulis* increases during the spawning season in response to a decrease in food availability [876]. However, some species seem to produce large eggs under poor feeding conditions, e.g. some poeciliid fishes [953], daphnids [412] and *Sancassania* mites [82]. Moreover, egg size can vary within a clutch [314, 822, 1216], according to geographical distribution [1065], with age [769] and race. Nonetheless, the pattern that the reserve density at birth $[E_b]$ equals the reserve density $[E]$ of the mother at egg formation generally holds; this not only removes a parameter, but also has the nice implication that the von Bertalanffy growth curve applies from birth on, at constant food

Table 2.1: The dimensionless scaled variables and parameters that are used to find the initial amount of scaled reserve.

$\tau = a\dot{k}_M$	$\tau_b = a_b\dot{k}_M$	$l = L/L_m$	$l_b = L_b/L_m$
$u_E = \frac{E}{g[E_m]L_m^3}$	$u_E^0 = \frac{E_0}{g[E_m]L_m^3}$	$u_H = \frac{E_H}{g[E_m]L_m^3}$	$u_H^b = \frac{E_H^b}{g[E_m]L_m^3}$
$e = gu_E/l^3$	$e_b = gu_E^b/l_b^3$	$e_H = gu_H/l^3$	$e_H^b = gu_H^b/l_b^3$
$x = \frac{g}{e+g}$	$x_b = \frac{g}{e_b+g}$	$\alpha = 3gx^{1/3}/l$	$\alpha_b = 3gx_b^{1/3}/l_b$
$y = \frac{xe_H}{1-\kappa}$	$y_b = \frac{x_b e_H^b}{1-\kappa} = gx_b v_H^b l_b^{-3}$	$v_H^b = \frac{u_H^b}{1-\kappa}$	$k = \dot{k}_J/\dot{k}_M$

density. Embryos don't feed, $f = 0$, and even embryos of endothemic species don't allocate to heating, $L_T = 0$.

The state variables (E_H, E, L) evolve from $(0, E_0, 0)$ at age $a = 0$ to $(E_H^b, [E_b]L_b^3, L_b)$ at age $a = a_b$, the age at birth. To find E_0, a_b, and L_b, given E_H^b and $[E_b]$ is a bit of a challenge, which has been overcome only recently [645].

For this purpose, it is most convenient to remove parameters by scaling to dimensionless quantities: (τ, u_E, l, u_H), see Table 2.1. The reformulated problem is now: find τ_b, l_b, u_E^0 given u_H^b, k, g, κ and $u_E^b = e_b l_b^3/g$, where e_b is the scaled reserve density at birth.

For the variable (τ, u_E, l, u_H) evolving from the value $(0, u_E^0, 0, 0)$ to the value $(\tau_b, u_E^b, l_b, u_H^b)$, the scaled model amounts to

$$\frac{d}{d\tau}u_E = -u_E l^2 \frac{g+l}{u_E + l^3} \tag{2.26}$$

$$\frac{d}{d\tau}l = \frac{1}{3}\frac{gu_E - l^4}{u_E + l^3} \tag{2.27}$$

$$\frac{d}{d\tau}u_H = (1-\kappa)u_E l^2 \frac{g+l}{u_E + l^3} - ku_H \tag{2.28}$$

or alternatively for variable (τ, e, l, e_H) evolving from the value $(0, \infty, 0, e_H^0)$ to the value $(\tau_b, e_b, l_b, e_H^b)$

$$\frac{d}{d\tau}e = -g\frac{e}{l} \tag{2.29}$$

$$\frac{d}{d\tau}l = \frac{g}{3}\frac{e-l}{e+g} \tag{2.30}$$

$$\frac{d}{d\tau}e_H = (1-\kappa)\frac{ge}{l}\frac{l+g}{e+g} - e_H\left(k + \frac{g}{l}\frac{e-l}{e+g}\right) \tag{2.31}$$

where the initial scaled maturity density $e_H^0 = (1-\kappa)g$ is such that $\frac{d}{d\tau}e_H(0) = 0$, otherwise $\frac{d}{d\tau}e_H(0) = \pm\infty$.

If $k = 1$, so $\dot{k}_J = \dot{k}_M$, we have $e_H(\tau) = e_H^0$ for all τ and $u_H(\tau) = (1-\kappa)l_b^3$. In other words: the maturity density remains constant, so maturity exceeds threshold values when structure exceeds threshold values. We then have the relationship for the structural volume at birth

$$V_b = L_b^3 = \frac{E_H^b/[E_m]}{(1-\kappa)g} \tag{2.32}$$

For $k > 1$, e_H is decreasing in (scaled) age, and for $k < 1$ increasing.

Figure 2.12 shows that data on embryo weight, yolk and respiration are in close agreement with model expectations. As is discussed later, {143}, respiration is taken to be proportional to the mobilisation rate. The two or three curves per species have been fitted simultaneously by Zonneveld [1294], and the total number of parameters is five excluding, or seven including, respiration. This is fewer than three parameters per curve and thus approaches a straight line for simplicity when measured this way. I have not found comparable data for plant seeds, but I expect a very similar pattern of development.

The examples are representative of the data collected in Table 2.2, which gives parameter estimates of some 40 species of snails, fish, amphibians, reptiles and birds. The model tends to underestimate embryo weight and respiration rate in the early phases of development. This is partly because of deviations in isomorphism, the contributions of extra-embryonic membranes (both in weight and in the mobilisation of energy reserve) and the loss of water content during development. The parameter estimates for altricial birds such as the parrot *Agapornis* should be treated with some reservation, because neglected acceleration caused by the temperature increase during development substantially affects the estimates, as discussed on {167}.

The values for the energy conductance $\dot{v}$, as given in Table 2.2, are in accordance with the average value for post-embryonic development, as given on {303}, which indicates that no major changes in energy parameters occur at birth. The maintenance rate coefficient $\dot{k}_M$ for reptiles and birds is about $0.08\,\mathrm{d}^{-1}$ at 30 °C, implying that the energy required to maintain tissue for 12 days at 30 °C is about equal to the energy necessary to synthesise the tissue from the reserve. The maintenance rate constant for freshwater species seems to be much higher, ranging from 0.3 to $2.3\,\mathrm{d}^{-1}$. Data from Smith [1079] on the rainbow trout *Oncorhynchus mykiss* result in $1.8\,\mathrm{d}^{-1}$ and Figure 2.10 gives $4.06\,\mathrm{d}^{-1}$ at 20 °C, which corresponds to some $8.5\,\mathrm{d}^{-1}$ at 30 °C (if it would survive that) for the waterflea *Daphnia magna*. The costs of osmosis might contribute to these high maintenance costs, as has been suggested on {43}, but for Ecdysozoa (to which *Daphnia* belongs), moulting might be costsly. The high value for *Oikopleura*, see Figure 2.19, probably relates to house production. Although information on parameter values is still sparse, it indicates that no (drastic) changes in these values occur at the transition from the embryonic to the juvenile state.

The general pattern of embryo development in eggs is characterised by unrestricted fast development during the first part of the incubation period (once the process has started) due to unlimited energy supply, at a rate that would be impossible to reach if the animal had to refill reserves by feeding. This period is followed by a retardation of development due to the increasing depletion of energy reserves. Apart from the reserves of the juvenile, the model works out very similar to that of Beer and Anderson [74] for salmonid embryos.

In view of the goodness of fit of the model in species that do not possess shells (see the turtle data), retardation is unlikely to be due to limitation of gas diffusion across the shell, as has been frequently suggested for birds [930]. The altricial and precocial modes of development have been classified as being basically different; the precocials show a plateau in respiration rates towards the end of the incubation period, whereas the altricials do not. Figure 2.13 shows that this difference can be traced back to the simple fact that

Figure 2.12: Yolk-free embryo weight (⋄), yolk weight (×) and respiration rate (+) during embryo development, and fits on the basis of the DEB model. Data sources are indicated.

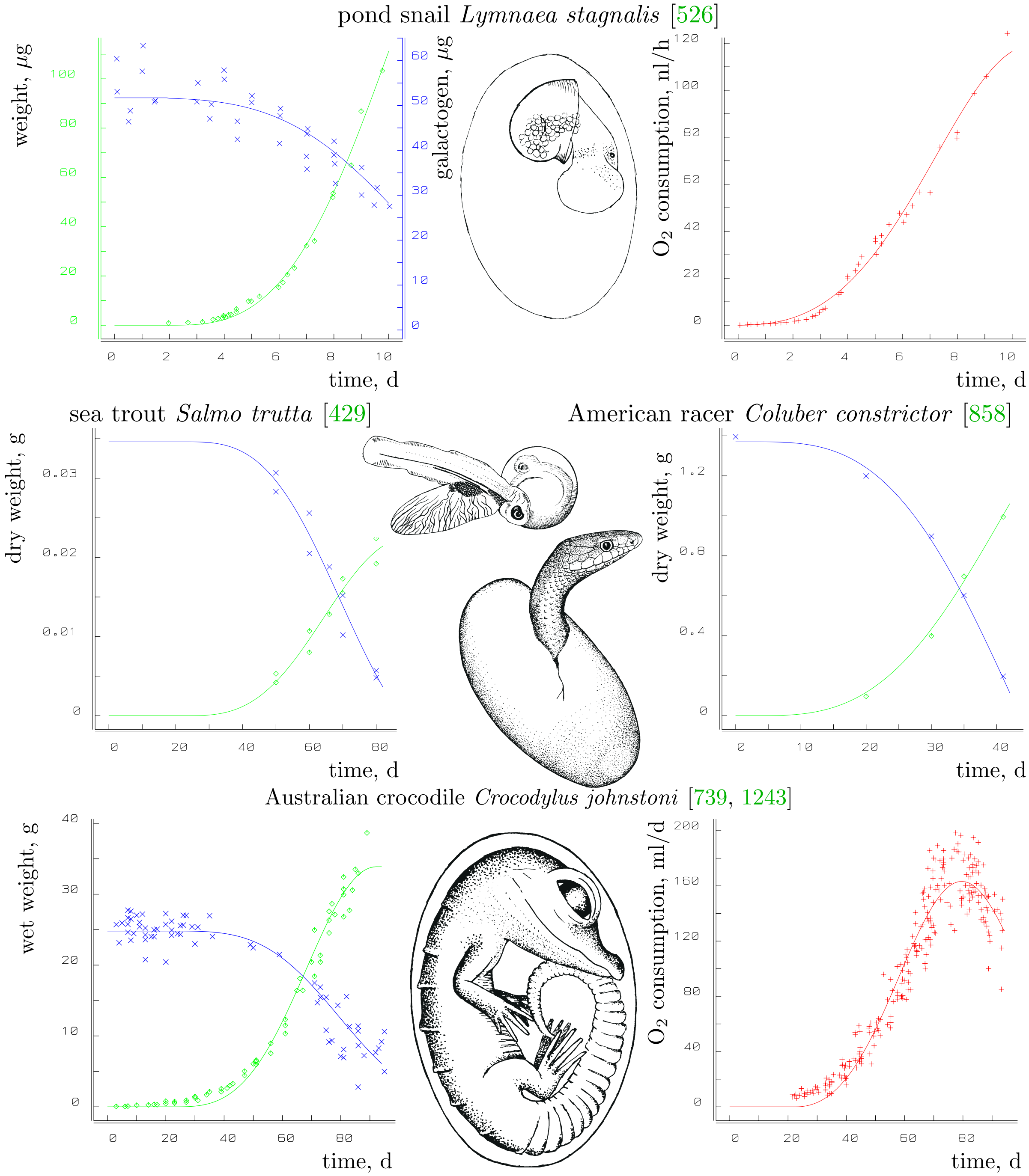
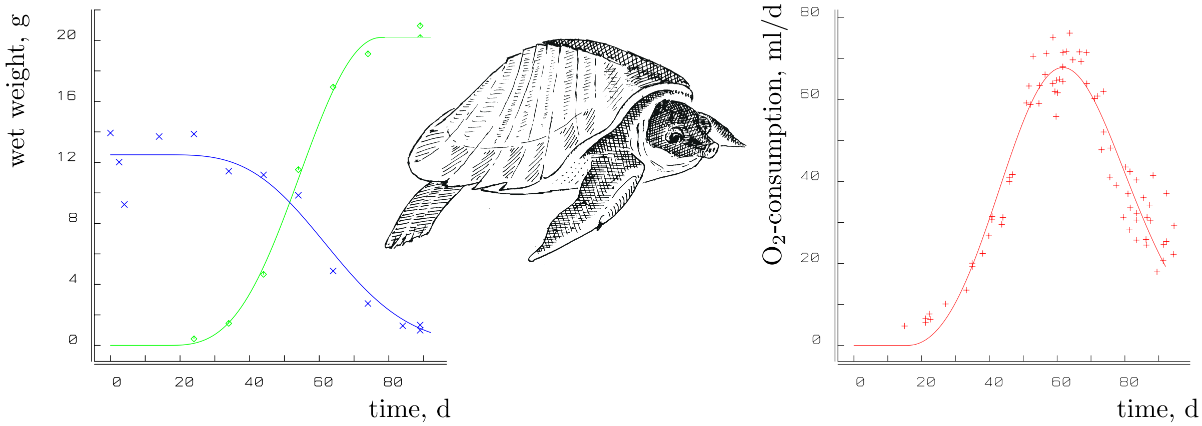


Figure 2.12 continued

New Guinea soft-shelled turtle *Carettochelys insculpta* [1224]



Laysan albatross *Diomedea immutabilis* [885]

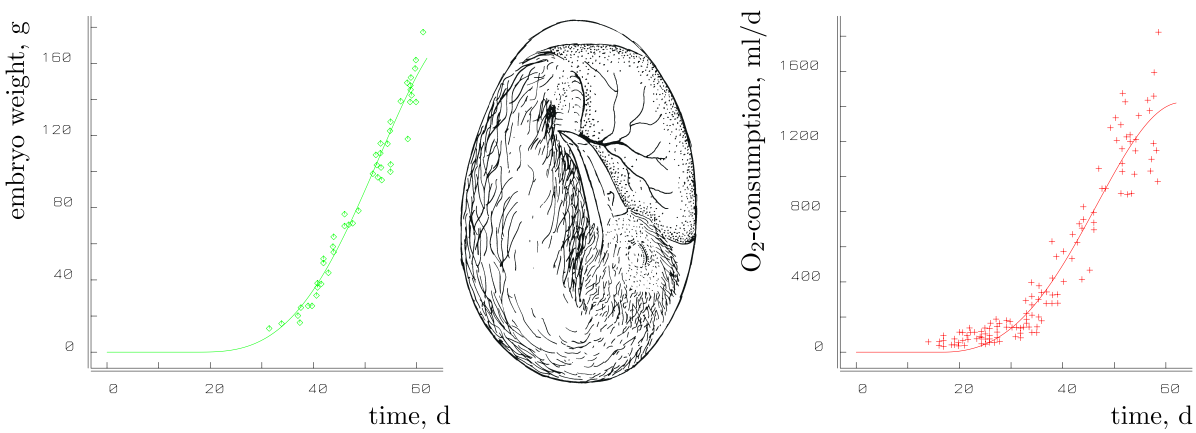


Table 2.2: Survey of re-analysed egg data, and parameter values standardised to a temperature of 30 °C, taken from [1294]. *1* P. J. Whitehead, pers. comm., 1989 ; *2* M. B. Thompson, pers. comm., 1989; 'galac.', stands for galactogen content.

species	temp. ° C	type of data	$\dot{v}_{30}$ mm d^{-1}	$\dot{k}_{M\,30}$ d^{-1}	E_b/E_0	reference
Lymnaea stagnalis	23	ED, galac, O	0.80	2.3	0.55	[526]
Salmo trutta	10	ED, YD	3.0	0.31	0.37	[429]
Rana pipiens	20	EW, O	2.5		0.87	[40]
Crocodylus johnstoni	30	EW, YW	1.9	0.060	0.31	[739]
	29, 31	O				[1243]
Crocodylus porosus	30	EW, YW	2.7	0.024	0.19	[1225]
	30	O				*1*
Alligator mississippiensis	30	EW, YW	2.7		0.34	[256]
	30	O				[1151]
Chelydra serpentina	29	ED, YD	1.9		0.35	[858]
	29	O				[398]
Carettochelys insculpta	30	EW, YW, O	1.9	0.040	0.08	[1224]
Emydura macquarii	30	EW, O	1.6	0.14	0.35	[1151]
Caretta caretta	28–30	EW, O	3.0		0.65	[4, 3]
Chelonia mydas	28–30	EW, O	3.0		0.57	[4, 3]
Amphibolurus barbatus	29	ED, YD	0.92	0.061	0.47	[859]
Coluber constrictor	29	ED, YD	1.4		0.69	[860]
Sphenodon punctatus	20	HW, O	0.85	0.062	0.25	*2*
Gallus domesticus	39	EW, O, C	3.2	0.039	0.34	[985]
Gallus domesticus	38	EW, C	3.4		0.52	[125]
Leipoa ocellata	34	EE, YE, O	1.7	0.031	0.55	[1198]
Pelicanus occidentalis	36.5	EW, O	3.2	0.10	0.77	[61]
Anous stolidus	35	EW, O	2.0	0.11	0.59	[886]
Anous tenuirostris	35	EW, O	1.8	0.20	0.59	[886]
Diomedea immutabilis	35	EW, O	2.5	0.069	0.57	[885]
Diomedea nigripes	35	EW, O	2.5	0.049	0.58	[885]
Puffinus pacificus	38	EW, O	0.92	0.084	0.61	[5]
Pterodroma hypoleuca	34	EW, O	1.9		0.20	[885]
Larus argentatus	38	EW, C	2.7	0.15	0.56	[293]
Gygis alba	35	EW, O	1.4		0.53	[884]
Anas platyrhynchos	37.5	EW	2.5	0.10	0.67	[920]
	37.5	O				[577]
Anser anser	37.5	EW	4.1	0.039	0.23	[984]
	37.5	O				[1196]
Coturnix coturnix	37.5	EW, O	1.7		0.49	[1196]
Agapornis personata	36	EW, O	0.8		0.79	[170]
Agapornis roseicollis	36	EW, O	0.84		0.81	[170]
Troglodytes aëdon	38	EW, O	1.4		0.82	[589]
Columba livia	38	EW	2.7		0.80	[581]
	37.5	O				[1196]

EW: Embryo Wet weight YW: Yolk Wet weight ED: Embryo Dry weight
EE: Embryo Energy content YE: Yolk Energy content YD: Yolk Dry weight
O: Dioxygen consumption rate C: Carbon dioxide prod. rate HW: Hatchling Wet weight

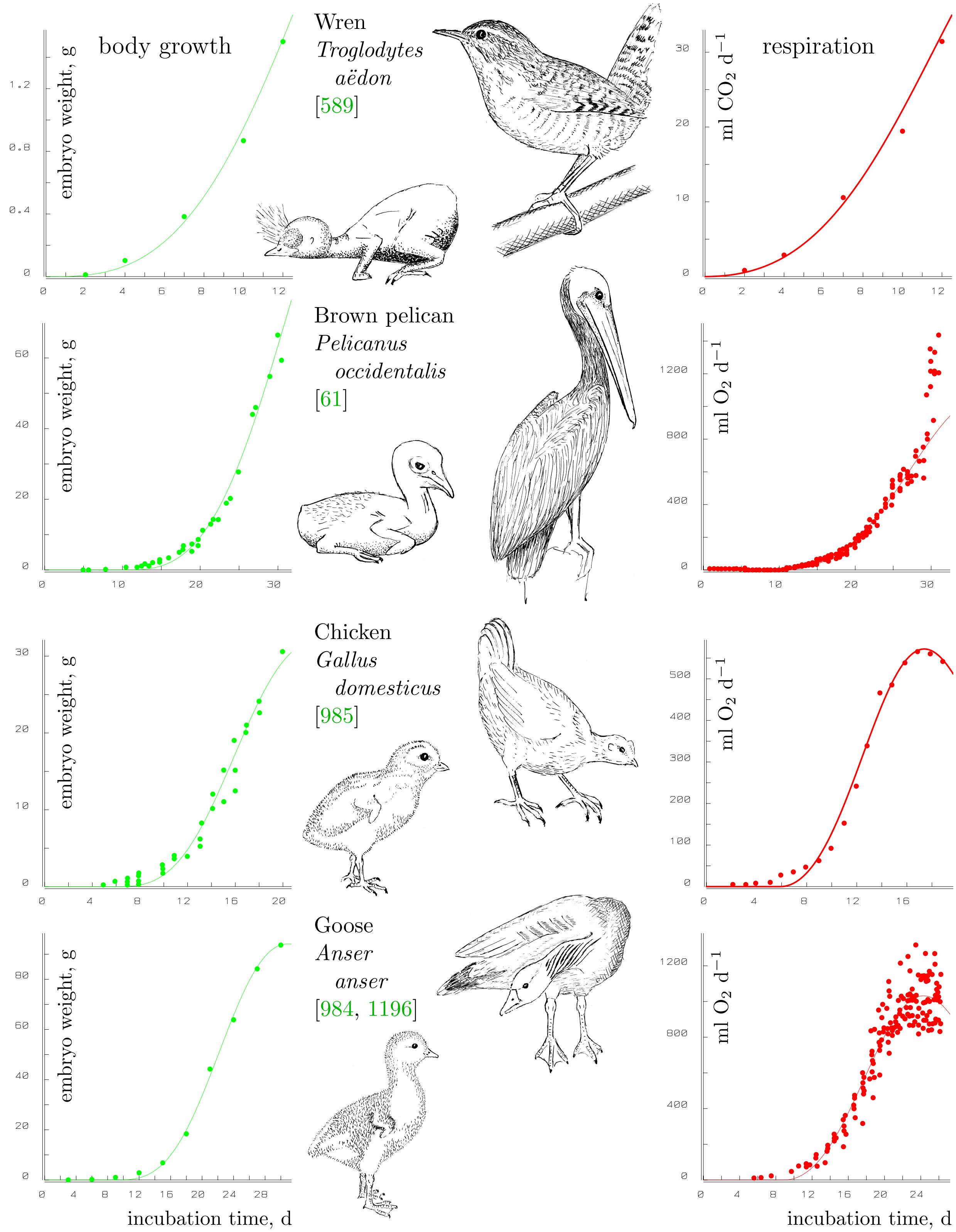


Figure 2.13: The embryonic development of altricial (wren, pelican) and precocial (chicken, goose) birds. Data from the sources indicated; fits are on the basis of the DEB model (parameters in Table 2.2). The underestimation of the initial development possibly relates to embryonic membranes. Pelican's high respiration rate just prior to hatching is attributed to internal pipping, which is not modelled. The drawings show hatchlings and adults.

altricial birds hatch relatively early. A frequently used argument for diffusion limitation is the strong negative correlation between diffusion rates across the egg shell and diffusion resistance, when different egg sizes are compared, ranging from hummingbirds to ostriches; the product of diffusion rate and resistance does not vary a lot. This correlation probably results from a minimisation of water loss by eggs. Data from fossil plants from Greenland show a dramatic drop in stomata frequency at the end of the Triassic (208 Ma ago), which has been linked to a steep rise of the atmospheric carbon dioxide concentration to three times the present values, possibly as a result of the activity of volcanoes that mark the breakup of Pangaea [75, p98]. The plants no longer needed many stomata, and reduced the number to reduce the water loss by evaporation.

To find τ_b, l_b, u_E^0, I first observe that from Table 2.1 and ODEs (2.26 – 2.27), we have

$$\frac{d}{d\tau}x = gx\frac{1-x}{l}; \quad \frac{d}{d\tau}l = \frac{g - xg - lx}{3}; \quad \frac{d}{d\tau}\alpha = \frac{x^{1/3}}{1-x}\frac{d}{d\tau}x \tag{2.33}$$

so

$$\alpha = 3g(u_E^0)^{-1/3} + B_x(\frac{4}{3}, 0) \tag{2.34}$$

for the incomplete beta function

$$B_x(\frac{4}{3}, 0) \equiv \int_0^x y^{1/3}(1-y)^{-1}\, dy \tag{2.35}$$
$$= \sqrt{3}\left(\arctan\frac{1+2x^{1/3}}{\sqrt{3}} - \arctan\frac{1}{\sqrt{3}}\right) + \frac{1}{2}\log(1 + x^{1/3} + x^{2/3}) - \log(1 - x^{1/3}) - 3x^{1/3}$$

Consequently we have

$$\alpha_b - \alpha = B_{x_b}(\frac{4}{3}, 0) - B_x(\frac{4}{3}, 0) \tag{2.36}$$
$$\frac{1}{l} = \frac{1}{l_b}\left(\frac{x_b}{x}\right)^{1/3} - \frac{B_{x_b}(\frac{4}{3},0) - B_x(\frac{4}{3},0)}{3gx^{1/3}} \tag{2.37}$$

We need this expression for $l(x)$ later in the derivation of l_b.

Scaled age at birth τ_b

The scaled age at birth τ_b follows from (2.33) and (2.37) by separation of variables and integration

$$\tau_b = 3\int_0^{x_b} \frac{dx}{(1-x)x^{2/3}(\alpha_b - B_{x_b}(\frac{4}{3},0) + B_x(\frac{4}{3},0))} \tag{2.38}$$

Notice that τ_b requires l_b in α_b, which is given below.

The parameter values in Figure 2.10 for *D. magna* imply an incubation time of 0.76 d at $f = 1$ till 0.80 d at $f = 0.5$. The eggs are deposited in the brood pouch just after moulting and develop there till birth just before the next moult, some 1.5 to 2 d later at 20 °C. This suggests a food-level-dependent diapause of around 0.5 d in *D. magna*.

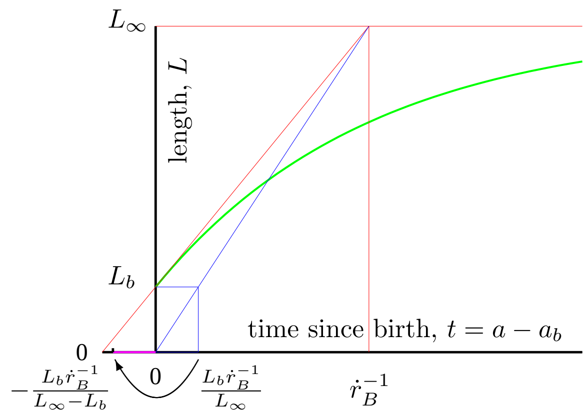


Figure 2.14: The von Bertalanffy growth curve $\frac{d}{dt}L = \dot{r}_B(L_\infty - L)$, with the graphical interpretation of the von Bertalanffy growth rate $\dot{r}_B$, and the maximum possible age at birth $\frac{L_b \dot{r}_B^{-1}}{L_\infty}$ in the context of DEB theory. The tangent line at $t = 0$ intersects the asymptote at time $\dot{r}_B^{-1}$; the line from the origin to this intersection point hits level L_b at time $\frac{L_b}{L_\infty \dot{r}_B}$, which is the maximum possible incubation time.

The age at birth simplifies for small g and large $\dot{k}_M$, while $\dot{r}_B = \frac{\dot{k}_M g}{3(e_b+g)}$ remains fixed [660]:

$$a_b = \frac{3}{\dot{k}_M} \int_0^{x_b} \frac{dx}{(1-x)x^{2/3}(3gx_b^{1/3} l_b^{-1} - B_{x_b}(\frac{4}{3}, 0) + B_x(\frac{4}{3}, 0))} \quad (2.39)$$

$$\overset{g, \dot{k}_M^{-1} \text{ small}}{\simeq} \frac{1}{3e_b \dot{r}_B} \int_0^{x_b} \frac{dx}{(1-x)x^{2/3} x_b^{1/3}(l_b^{-1} - (x_b^{4/3} - x^{4/3})/(4e_b))}$$

$$\overset{g, \dot{k}_M^{-1} \text{ very small}}{\simeq} \frac{l_b}{3e_b \dot{r}_B} \int_0^{x_b} \frac{dx}{(1-x)x^{2/3} x_b^{1/3}} \overset{g, \dot{k}_M^{-1} \to 0}{=} \frac{l_b}{e_b \dot{r}_B} \overset{e_b = f}{=} \frac{3l_b}{\dot{k}_M g}\left(1 + \frac{g}{f}\right) \quad (2.40)$$

where $x_b \equiv \frac{g}{e_b+g}$. The significance of this result is in the fact that for fixed $\dot{r}_B$, L_b and L_∞, $g \to 0$ while $\dot{k}_M \to \infty$ if a_b is running from 0 to this upper boundary. See Figure 2.14 for a graphical interpretation.

The chameleon *Furcifer labordi* is reported to live 8-9 months as egg and only 4-5 months as juvenile plus adult [576]. The males grow from some 3 cm to 10.2 cm (snout-to-vent length) with a von Bertalanffy growth rate of $0.035\,\mathrm{d}^{-1}$, and the females from 2.6 cm to 7.8 cm with a von Bertalanffy growth rate of $0.04\,\mathrm{d}^{-1}$. The maximum age at birth that is consistent with DEB theory is thus some 8.3 d, rather than the observed 8-9 months. The eggs are buried in the soil, where it might be some 10 °C cooler than during post-embryonic growth. Even after correction for this difference, this suggests that this species is on a metabolic hold for most of the time.

The foetal special case, where $a_b = 3L_b/\dot{v} = \frac{3l_b}{\dot{k}_M g}$, represents a lower boundary for the age at birth (of eggs). So the possible range for a_b is

$$\frac{3l_b}{\dot{k}_M g} < a_b < \frac{3l_b}{\dot{k}_M g}(1 + g/e_b) \quad \text{or} \quad 1 < a_b \frac{\dot{k}_M g}{3l_b} < 1 + g/e_b \quad (2.41)$$

Scaled initial amount of reserve u_E^0

The scaled initial amount of reserve u_E^0 directly follows from (2.34) for $x = x_b$ and $\alpha = \alpha_b$

$$u_E^0 = \left(\frac{3g}{\alpha_b - B_{x_b}\left(\frac{4}{3}, 0\right)}\right)^3 \quad \text{so } E_0 = u_E^0 g [E_m] L_m^3 \tag{2.42}$$

which again requires l_b in α_b.

The parameter values in Figure 2.10 for *D. magna* imply a scaled initial amount of reserve of $U_E^0 = 0.04\,\mathrm{mm^2d}$ at $f = 1$ and $0.0246\,\mathrm{mm^2d}$ at $f = 0.5$ at 20 °C. The scaled amount at birth equals $U_E^b = f[E_m]L_b^3/\{\dot{p}_{Am}\} = 0.0313$ and $0.0156\,\mathrm{mm^2d}$, respectively. So a fraction of 0.79 and 0.63 for $f = 1$ and 0.5, respectively, of the initial reserve is still left at birth. Figure 2.10 has, however, no data on embryo development or reserve, which demonstrates the strength of DEB theory.

Data on embryo development, Table 2.2, also shows that about half of the reserves are used during embryonic development. The deviating values for altricial birds are artifacts, caused by the above–mentioned acceleration of development by increasing temperatures. Congdon *et al.* [217] observed that the turtles *Chrysemus picta* and *Emydoidea blandingi* have 0.38 of the initial reserves at birth. Respiration measurements on seabirds by Pettit *et al.* [887] indicate values that are somewhat above the ones reported in the table. The extremely small value for the softshelled turtle, see also Figure 2.12, relates to the fact that these turtles wait for the right conditions to hatch, after which they have to run the gauntlet as a cohort at night from the beach to the water, where a variety of predators wait for them.

Scaled length at birth l_b

For the variable (τ, e_H) evolving from the value $(0, e_H^0)$ to the value (τ_b, e_H^b) we have

$$\frac{d}{d\tau} e_H = (1-\kappa) g (1-x) \left(\frac{g}{l(x)} + 1\right) - e_H \left(k - x + g\frac{1-x}{l(x)}\right) \tag{2.43}$$

Now consider the variable (x, e_H) evolving from the value $(0, e_H^0)$ to the value (x_b, e_H^b) or the variable (x, y) evolving from the value $(0, 0)$ to the value (x_b, y_b):

$$\begin{aligned} \frac{d}{dx} e_H &= \frac{e_H^0}{x}\left(\frac{l(x)}{g} + 1\right) - \frac{e_H}{x}\left(\frac{k-x}{1-x}\frac{l(x)}{g} + 1\right) \quad \text{for } e_H^0 = e_H(0) = (1-\kappa)g \\ \frac{d}{dx} y &= r(x) - y s(x) \quad \text{for } r(x) = g + l(x); \quad s(x) = \frac{k-x}{1-x}\frac{l(x)}{gx} \end{aligned} \tag{2.44}$$

where $l(x)$ is given in (2.37). The ODE for y can be solved to

$$y(x) = v(x) \int_0^x \frac{r(x_1)}{v(x_1)} dx_1 \quad \text{with } v(x) = \exp(-\int_0^x s(x_1)\, dx_1) \tag{2.45}$$

The quantity l_b must be solved from $y_b = y(x_b) = g x_b v_H^b l_b^{-3}$, see Table 2.1. So we need to find the root of t as function of l_b with

$$t(l_b) = \frac{x_b g v_H^b}{v(x_b) l_b^3} - \int_0^{x_b} \frac{r(x)}{v(x)} dx = 0 \tag{2.46}$$

From this equation it becomes clear that the parameters κ and u_H^b affect l_b only via $v_H^b = \frac{u_H^b}{1-\kappa}$; a conclusion that is more difficult to obtain using the ODE for the scaled maturity density e_H rather than that for abstract variable y. Notice that the solution of l_b (and that of u_E^0 and τ_b) for the boundary value problem for the ODE for (u_E, l, e_H) as given in (2.26–2.28) depends on the four parameters g, k, v_H^b and e_b only. The solution for l_b must be substituted into (2.42) to obtain u_E^0 and in (2.38) to obtain τ_b; the scaled reserve at birth is $u_E^b = e_b l_b^3 / g$.

The parameter values in Figure 2.10 for *D. magna* imply a birth length of 0.686 and 0.685 mm at $f = 1$ and 0.5, respectively. These values hardly differ much because $\dot{k}_J$ is close to $\dot{k}_M$.

Special case $e \to \infty$: foetal development

Foetal development differs from that in eggs in that energy reserves are supplied continuously via the placenta. The feeding and digestion processes are not involved. Otherwise, foetal development is taken to be identical to egg development, with initial reserves that can be taken to be infinitely large, for practical purposes. At birth, the neonate receives an amount of reserves from the mother, such that the reserve density of the neonate equals that of the mother. So the approximation $[E] \to \infty$ or $e \to \infty$ for the foetus can be made for the whole gestation period, because the foetus lives on the reserves of the mother. In other words: unlike eggs, the development of foetuses is not restricted by energy reserves. Initially the egg and foetus develop in the same way, but the foetus keeps developing at a rate not restricted by the amount of reserves till the end of the gestation time, while the development of the egg becomes retarded, due to depletion of the reserves.

The special case $e \to \infty$ makes that $\frac{d}{d\tau} l = g/3$, or $l(\tau) = g\tau/3$. The foetal structural volume thus behaves as

$$\frac{d}{dt} V = \dot{v} V^{2/3} \quad \text{so } V(t) = (\dot{v} t/3)^3 \tag{2.47}$$

This growth curve was proposed by Huggett and Widdas [535] in 1951. Payne and Wheeler [873] explained it by assuming that the growth rate is determined by the rate at which nutrients are supplied to the foetus across a surface that remains in proportion to the total surface area of the foetus itself. This is consistent with the DEB model, which gives the energy interpretation of the single parameter. The graph of foetal weight against age resembles an exponential growth curve, but in fact it is less steep; the model has the property that subsequent weight doubling times increase by a factor $2^{1/3} = 1.26$, while there is no increase in the case of exponential growth.

The fit is again excellent; see Figure 2.15. It is representative for the data collected in Table 2.3 taken from [1294]. A time lag for the start of foetal growth has to be incorporated, and this diapause may be related to the development of the placenta, which possibly

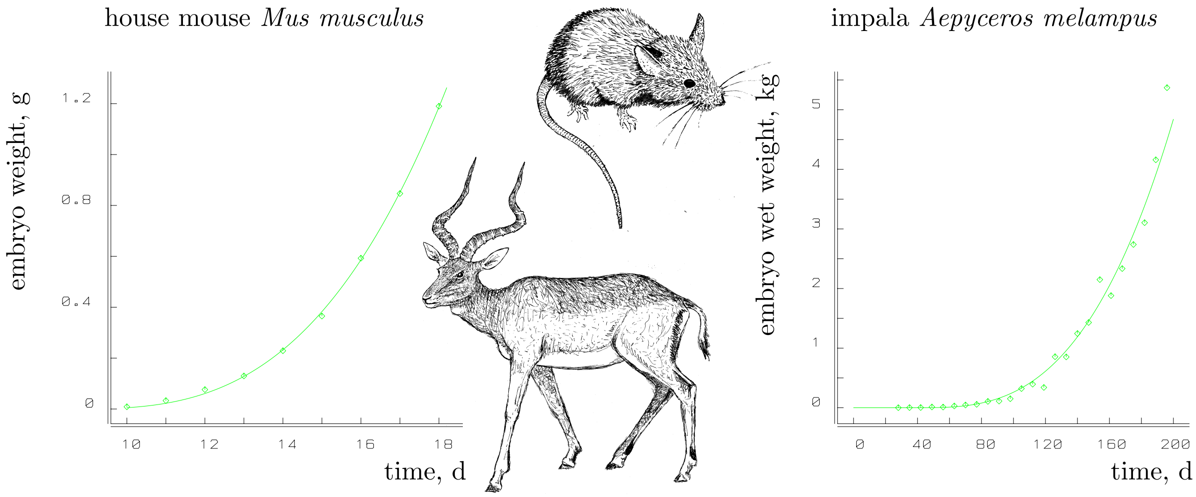


Figure 2.15: Foetal weight development in mammals, see Figure 6.2. Parameters are given in Table 2.3.

depends on body volume as well. The long diapause for the grey seal *Halichoerus* probably relates to timing with the seasons to ensure adequate food supply for the developing juvenile. Variations in weight at birth are primarily due to variations in gestation period, not in foetal growth rate. For comparative purposes, energy conductance $\dot{v}$ is converted to 30 °C, on the assumption that the Arrhenius temperature, T_A, is 10 200 K and the body temperature is 37 °C for all mammals in the table. This is a rather crude conversion because the cat, for instance, has a body temperature of 38.6 °C. Weights were converted to volumes using a specific density of $[W_w] = 1$ g cm^{-3}.

One might expect that precocial development is rapid, resulting in advanced development at birth and, therefore, comes with a high value for the energy conductance. The values collected in Table 2.3, however, do not seem to have an obvious relationship with altricial–precocial rankings. The precocial guinea-pig and alpaca as well as the altricial human have relatively low values for the energy conductance. The altricial–precocial ranking seems to relate only to the relative volume at birth V_b/V_m.

We further have

$$\frac{d}{d\tau}u_H = (1-\kappa)l^2(g+l) - ku_H \tag{2.48}$$

$$\frac{u_H(\tau)3^3k^4}{g^3(1-\kappa)} = 6k\tau - 3k^2\tau(2+\tau) + k^3\tau^2(3+\tau) - 6(1-k)\left(1-\exp(-k\tau)\right) \tag{2.49}$$

The equation $u_H(\tau_b) = u_H^b$ has to be solved numerically for τ_b, but for $k = 1$ we have $u_H^b = (1-\kappa)3^{-3}g^3\tau_b^3 = (1-\kappa)l_b^3$. The solution of this equation is stable and fast; the resulting scaled length at birth $l_b = g\tau_b/3$ can be used to start the Newton Raphson procedure. This start is preferable if k is substantially different from 1. From $l_b < 1$, so $\tau_b < 3/g$, we can derive the constraint

$$\frac{k^2u_H^b}{1-\kappa} < k + g(k-1) + g^3\frac{k-1}{k^2}\,\frac{1-3/g-\exp(-3k/g)}{9/2} \tag{2.50}$$

Table 2.3: The estimated diapause t_0, age at birth a_b, scaled energy conductance $\dot{v}^* = \dot{v}(1+\omega_w)^{1/3}$, birth weight W_b for mammals; specific density $d_V = 1\,\mathrm{g\,cm^{-3}}$, $e = 1$.

species	t_0 d	a_b d	$\dot{v}^*$ cm d^{-1}	W_b g	reference
Homo sapiens	26.8	259	0.180	3750	[1228]
Oryctolagus cuniculus	10.7	19.2	0.560	46.2	[694]
Lepus americanus	13.1	21.1	0.573	66	[127]
Cavia porcellus	15.7	48.8	0.269	84	[291]
	19.5	48.5	0.239	101	[540]
Cricetus auratus	9.29	5.8	0.570	1.43	[924]
Mus musculus	8.2	11.8	0.278	1.2	[727]
Rattus norvegicus					
wistar	11.4	7.7	0.487	2	[346]
	11.8	9.3	0.525	4.3	[535]
albino	12.4	9.5	0.542	5	[355]
Clethrionomys glareolus	8.29	10.4	0.374	2.2	[224]
Aepyceros melampus	39.4	166	0.316	5370	[331]
Odocoileus virginianus	34.9	155	0.296	3577	[975]
Dama dama	9.94	155	0.345	56400	[35]
Cervus canadensis	24.9	217	0.336	14500	[810]
Lama pacus	7.47	90.7	0.120	47.7	[347]
Ovis aries					
Welsh	43.9	105.8	0.482	4913	[535]
	33.3	111	0.433	4120	[213]
Karakul	31.0	116	0.426	4445	[306]
	27.5	121.8	0.403	4384	[567]
Capra hircus	24.3	126.3	0.339	2910	[317]
	31.3	117.3	0.365	2910	[59]
Bos taurus	59.5	204	0.475	33750	[1259]
Equus caballus	37.0	348	0.370	79200	[788]
Sus scrofa	4.73	129	0.266	1500	[1217]
Yorkshire	5.49	121	0.283	1500	[1176]
large white	23.6	90	0.383	1500	[906]
Essex	14.1	107	0.321	1500	[906]
Felix catus	18.8	39.8	0.371	119.5	[223]
Pipistrellus pipistrellus					
1978	9.95	14.2	0.237	1.4	[925]
1979	13.7	18.5	0.181	1.4	[925]
Halichoerus grypus	145	165	0.375	8732	[500]

For $u_E^b = u_E(\tau_b)$, the cost for a foetus amounts to

$$u_E^0 = u_E^b + \kappa l_b^3 + u_H^b + \int_0^{\tau_b} (\kappa l^3(\tau) + k u_H(\tau))\, d\tau = u_E^b + l_b^3 + \frac{3}{4}\frac{l_b^4}{g}; \quad E_0 = u_E^0 g [E_m] L_m^3 \quad (2.51)$$

where the five terms correspond with the costs of reserve, structure, maturity, somatic and maturity maintenance, respectively. The second equality follows from the structure of DEB theory: the investment in maturity plus maturity maintenance equals $\frac{1-\kappa}{\kappa}$ times the investment in structure plus somatic maintenance and $l(\tau) = g\tau/3$. The foetal cost of a foetus is somewhat smaller than that of an egg (2.42).

Foetal development obviously affects the energetics of the mother. This is discussed at {275}.

2.6.3 States at puberty

Puberty occurs as soon as $E_H = E_H^p$, or $U_H = U_H^p = E_H^p/\{\dot{p}_{Am}\} = U_H^p$ or $u_H = u_H^p = \frac{E_H^p}{g[E_m]L_m^3}$. If $\dot{k}_J = \dot{k}_M$, and so maturity density is constant, and $L_T = 0$, we have

$$V_p = \frac{E_H^p/[E_m]}{(1-\kappa)g} \quad (2.52)$$

Otherwise the structural volume at puberty $V_p = L_p^3$ can vary with food density history and even more than V_b can. It must be found numerically from integration of $\frac{d}{dt}V$ till $E_H(t) = E_H^p$.

Generally little can be said about age and length at puberty, but if food density X, and so scaled functional response f, remains constant, (2.23), (2.22) and (2.43) show that age and scaled length at puberty amount to

$$a_p = a_b + \frac{1}{\dot{r}_B} \ln \frac{L_\infty - L_b}{L_\infty - L_p} \quad (2.53)$$

$$l_p = l(v_H^p) \quad \text{with} \quad \frac{d}{dv_H} l = \frac{(f - l - l_T)g/3}{f l^2 (g + l) - k v_H (g + f)} \quad \text{and} \quad l(v_H^b) = l_b \quad (2.54)$$

where $k = \dot{k}_J/\dot{k}_M$ as before. This differential equation has to be integrated numerically and results in a satiating function $l_p(f)$ if $k < 1$. We must have that $f > l_p + l_T$ to allow for adolescence.

The parameter values in Figure 2.10 for *D. magna* imply a length at puberty of 2.46 and 2.45 mm at $f = 1$ and 0.5, respectively. These values hardly differ because $\dot{k}_J$ is close to $\dot{k}_M$.

2.6.4 Reduction of the initial amount of reserve

The embryonic period is elongated if the initial amount of reserve is reduced, while the cumulative energy investment to complete the embryonic stage is the same. The mechanism is that reducing the amount of reserve reduces the mobilisation of reserve, so it takes longer

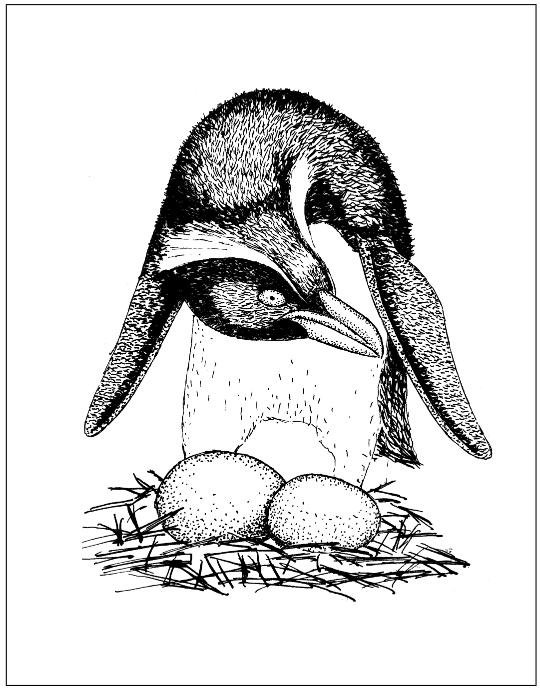


Figure 2.16: Egg dimorphism occurs as standard in crested penguins (genus *Eudyptes*). The small egg is laid first, but it hatches later than the big one, which is 1.5 times as heavy. The DEB theory explains why the large egg requires a shorter incubation period. The illustration shows the Snares crested penguin *E. atratus*.

to reach a certain maturity level. Large eggs, so large initial energy supplies, result in short incubation times if eggs of one species are compared. Crested penguins, *Eudyptes*, are known for egg dimorphism [1216]; see Figure 2.16. They first lay a small egg and, some days later a 1.5 times bigger one. As predicted by the DEB model, the bigger one hatches first, if fertile, in which case the parents cease incubating the smaller egg, because they are only able to raise one chick. They continue to incubate the small egg only if the big one fails to hatch. This is probably an adaptation to the high frequency of unfertilised eggs or other causes of loss of eggs (aggression [1216]), which occurs in this species.

Incubation periods only decrease for increasing egg size if all other parameters are constant. The incubation period is found to increase with egg size in some beetle species, lizards and marine invertebrates [210, 327, 1065]. In these cases, however, the structural biomass at hatching also increases with egg size. This is again consistent with the DEB theory, although the theory does not explain the variation in egg sizes.

Hart [465] studied the effect of separation of the embryonic cells of the sea urchin *Strongylocentrotus droebachiensis* in the two-cell stage on the energetics of larval development. Both the size and the feeding capacity of the resulting larva were reduced by about one-half, but the time to metamorphosis is about the same (7 d at 8–13 °C). The maximum clearance rate of dwarf and normal larvae was found to be the same function of the ciliated band length. Larvae fed a smaller ration had longer larval periods, but food ration hardly affected size at metamorphosis. Egg size affected juvenile test diameter only slightly.

Armadillos typically separate cells in the four-cell stage of the embryo, giving birth to four identical offspring. Humans rarely do this successfully, then giving birth to four babies of about 1 kg each, rather than the typical 3 kg. In terms of an effect on length this reduction amounts to a factor $(1/3)^{1/3} = 0.69$. The human growth curve fits the von

Bertalanffy curve very well, with a von Bertalanffy growth rate of $\dot{r}_B = \frac{\dot{k}_M g}{3(e+g)} = 0.123\ \mathrm{a}^{-1}$, see {282}. We can safely assume that the scaled reserve density was close to its maximum $e = 1$ for the post-embryonic stages. Moreover the age at birth is $a_b = \frac{3l_b}{g\dot{k}_M} = 0.75$ a for humans. If we take a typical maximum adult weight of 70 kg, then the scaled length at birth equals $l_b = (3/70)^{1/3} = 0.35$. So the energy investment ratio equals $g = \frac{l_b}{a_b\dot{r}_B} - 1 = 2.79$, the somatic maintenance rate coefficient $\dot{k}_M = \frac{3l_b}{ga_b} = 0.5\ \mathrm{a}^{-1}$ and the scaled age at birth $\tau_b = a_b\dot{k}_M = 0.375$. With these values for g, e_b and l_b, the scaled cost amounts to $u_E^0 = 0.062$ from (2.51). In the case of 4 babies with a length reduced by a factor 0.69, the scaled cost per baby equals $u_E^0 = 0.02$, so summed over the 4 babies this is 1.3 times the amount of a single baby; not a surprising result, in view of the 4 kg of babies relative to the 3 kg for a single baby.

If the separation of the cells affects the required cumulative investment in development, however, other predictions result. It is then quite well possible that incubation is hardly affected, while size at birth is. The Standard DEB model correctly predicts that a reduction of the feeding level elongates the larval period and hardly affects size at puberty for a particular relationship between the somatic and maturity maintenance costs.

As discussed at {292}, the reserve density capacity $[E_m] = \{\dot{p}_{Am}\}/\dot{v}$ scales with maximum structural length of a species. So species with a larger ultimate body size tend to have a relatively larger reserve capacity. It turned out that for the combination of parameter values as found for *D. magna* we have to apply a zoom factor of at least $z = 1.87$ to arrive at a minimum maximum body size for which cell separation might be successful.

For $k > 1$, the structural volume at birth increases after halving, and decreases for $k < 1$. Since reserve contributes to weight, the weight at birth is close to half of the original weight at birth, irrespective of the value of k. The age of the two-cell stage is probably smaller than $\tau_b/3$, but the results are very similar.

The removal of an amount of reserve at the start of the development, as is frequently done [351, 554, 555, 808, 1065], elongates the incubation time (as observed in the gypsy moth [992]), and reduces the reserve at hatching. This experiment simulates the natural situation, where the nutritional status of the mother affects to initial amount of reserve. The pattern is rather similar to that of the separation of cells at an early stage, because reductions of structure and maturity at an early stage have little effect. The initial amount of reserve is a U-shaped function of the reserve at birth. The right branch is explained by the larger amount of reserve at birth, the left branch by the larger age at birth, which comes with larger cumulative somatic maintenance requirements.

The size of neonates of trout and salmon was found to increase with initial egg size [314, 477], suggesting that $k < 1$ for salmonids. This also applies to the emu *Dromaius novaehollandiae* [305], and probably represents a general pattern.

2.7 Reproduction: excretion of wrapped reserve

Organisms can achieve an increase in numbers in many ways. Sea anemones can split off foot tissue that can grow into a new individual. This is not unlike the strategy of budding yeasts. Colonial species usually have several ways of propagating. Fungi have intricate sexual reproduction patterns involving more than two sexes. Under harsh conditions some animals can switch from parthenogenetic to sexual reproduction, others develop spores or other resting phases. It would not be difficult to fill a book with descriptions of all the possibilities. The standard DEB model assumes propagation via eggs; dividing organisms don't have an embryo or adult stage, and divide when the maturity exceeds a threshold. This resets maturity and reduces the amount of structure and reserve.

Energy allocation to reproduction equals

$$\dot{p}_R = (1-\kappa)\dot{p}_C - \dot{k}_J E_H^p \tag{2.55}$$

see (2.18), where E_H is now replaced by the constant E_H^p, because the flux to maturity is redirected to reproduction at puberty, which means that maturity does not change in adults. The costs of an egg E_0 or a foetus is given in (2.42) or (2.51), so the mean reproduction rate $\dot{R}$ in terms of number of eggs per time equals

$$\dot{R} = \frac{\kappa_R \dot{p}_R}{E_0} = \frac{\kappa_R \dot{k}_M}{v_E^0}\left(\frac{el^2}{e+g}(g+l_T+l) - kv_H^p\right) \tag{2.56}$$

for $v_E^0 = \frac{u_E^0}{1-\kappa}$, see Table 2.1.

At constant food density, where $e = f$, the reproduction rate is, according to (2.56), proportional to

$$\dot{R} \propto L^2 + \frac{\dot{k}_M}{\dot{v}} L^3 - L_R^2 \tag{2.57}$$

where L_R is just a constant, which depends on the nutritional status. Comparison of reproduction rates for different body sizes thus involves three compound parameters, i.e. the proportionality constant, $\dot{k}_M/\dot{v}$ and L_R, if all individuals experience the same food density for a long enough time. Figure 2.17 shows that this relationship is realistic, but that the notorious scatter for reproduction data is so large that access to the parameter $\dot{k}_M/\dot{v}$ is poor. The fits are based on guestimates for the maintenance rate coefficient, $\dot{k}_M = 0.011\ \mathrm{d}^{-1}$, and the energy conductance, $\dot{v} = 0.433\ \mathrm{mm\,d}^{-1}$ at 20 °C. The main reason for the substantial scatter in reproduction data is that they are usually collected from the field, where food densities are not constant, and where spatial heterogeneities, social interactions, etc., are common.

The reproduction rate of spirorbid polychaetes has been found to be roughly proportional to body weight [497]. On the assumption by Strathmann and Strathmann [1122] that reproduction rate is proportional to ovary size and that ovary size is proportional to body size (an argument that rests on isomorphy), the reproduction rate is also expected to be proportional to body weight. They observed that reproduction rate tends to scale with body weight to the power somewhat less than one for several other marine invertebrate species, and used their observation to identify a constraint on body size for brooding inside the body cavity. The DEB theory gives no direct support for this constraint; an allometric regression of reproduction rate against body weight would result in a scaling parameter between 2/3 and 1, probably close to 1, depending on parameter values.

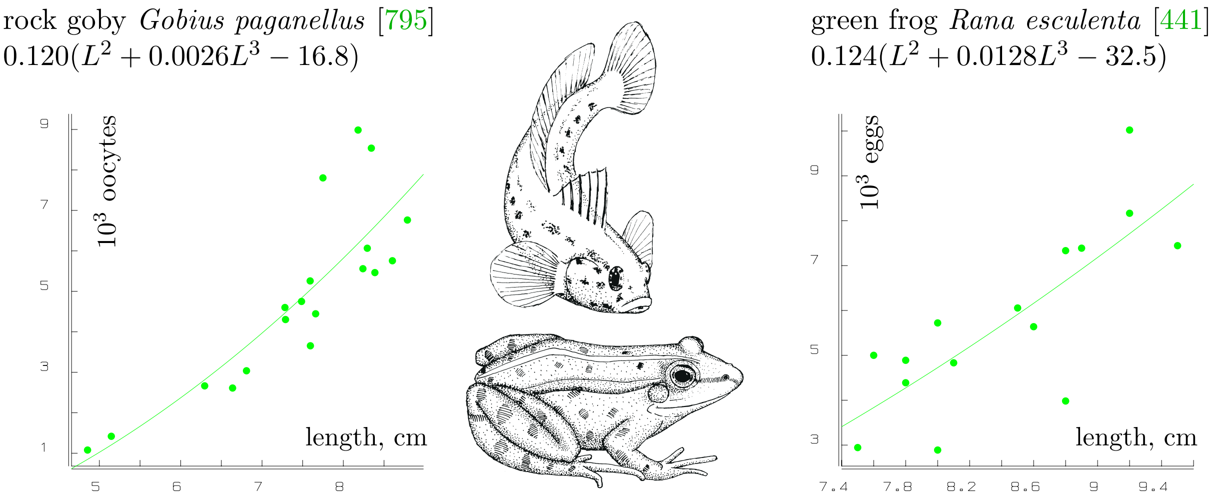


Figure 2.17: Clutch size, as a measure for reproduction rate, as a function of body length L for two randomly selected species. The data sources and DEB-based curves are indicated. The parameter that is multiplied by L^3 in both fits has been guestimated on the basis of common values for the maintenance rate coefficient and the energy conductance, with a shape coefficient of $\delta_{\mathcal{M}} = 0.1$ for the goby and of $\delta_{\mathcal{M}} = 0.5$ for the frog. Both other parameter values represent least-squares estimates.

The maximum (mean) reproduction rate for an individual of maximum volume, i.e. $l = 1 - l_T$, amounts to

$$\dot{R}_m = \kappa_R \dot{k}_M \frac{(1 - l_T)^2 - k v_H^p}{v_E^0} \tag{2.58}$$

with $v_E^0 = \frac{u_E^0}{1-\kappa}$, where u_E^0 is given in (2.42) for eggs and in (2.51) for foetuses.

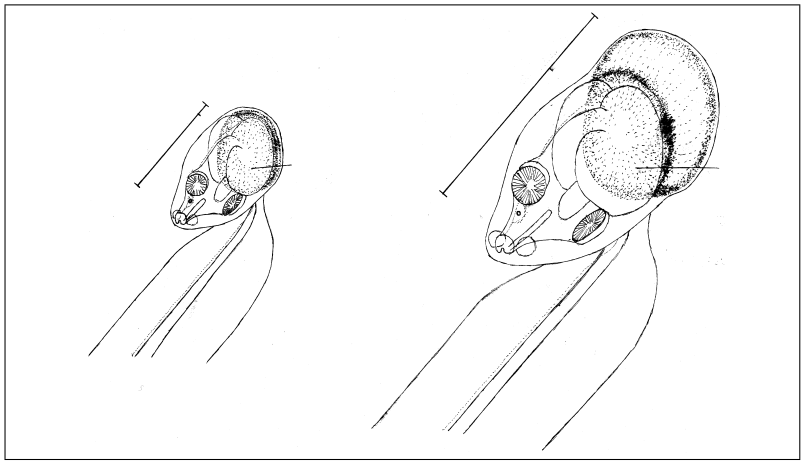
Under conditions of prolonged starvation, organisms can deviate from the standard reproduction allocation, as is discussed on {113}.

2.7.1 Cumulative reproduction

Oikopleura sports a heroic way of reproduction which leads to instant death. During its week-long life at 20 °C and abundant food, it accumulates energy for reproduction which is deposited at the posterior end of the trunk; see Figure 2.18. This allows an easy test of the allocation rule against experimental data. Except for this accumulation of material for reproduction, the animal remains isomorphic. The total length of the trunk, L_t, including the gonads, can be partitioned into the true trunk length, L, and the length of the gonads, L_R. Since the reproduction material is deposited on a surface area of the trunk, the length of the gonads is about proportional to the accumulated investment of energy in reproduction divided by the squared true trunk length. Fenaux and Gorsky [341] measured both the true and the total trunk length under laboratory conditions. This allows us to test the consequences of the DEB theory for reproduction.

Let $e_R(t_1, t_2)$ denote the cumulative investment of energy in reproduction between t_1 and t_2, as a fraction of the maximum energy reserves $[E_m]V_m$. From Table 2.5 (see below)

Figure 2.18: The larvacean *Oikopleura* grows isomorphically; during its short life it accumulates reproductive material at the posterior end of the trunk. The energy interpretation of data on total trunk lengths should take account of this. Larvaceans of the genus *Oikopleura* are an important component of the zooplankton of all seas and oceans and have an impact as algal grazers comparable with that of copepods.



we know that this investment amounts for adults to

$$e_R(t_1, t_2) = \kappa_R g \dot{k}_M \int_{t_1}^{t_2} \left((1-\kappa) \frac{g + l(t) + l_T}{g + e(t)} e(t) l^2(t) - k u_H^p \right) dt \qquad (2.59)$$

Oikopleura has a non-feeding larval stage and starts investing in reproduction as soon as it starts feeding, so $E_H^b = E_H^p$. From an energetic point of view, it thus lacks a juvenile stage, and the larva should be classified as an embryo. The total trunk length then amounts to $L_t(t) = L(t) + V_R e_R(0,t)/L^2(t)$. The volume V_R is a constant that converts the scaled cumulative reproductive energy per squared trunk length into the contribution to the total length. With abundant food, the true trunk length follows the von Bertalanffy growth curve (2.23) and $e(t) = 1$. If the data set $\{t_i, L(t_i), L_t(t_i)\}_{i=1}^n$ is available, the five parameters L_b, L_m, $\dot{k}_M$, g and $V_R\kappa_R$ can be estimated in principle. Dry weight relates to trunk length and reproductive energy as $W_d(t) = [W_{Ld}]L^3(t) + W_{Rd}e_R(0,t)$, where the two coefficients give the contribution of cubed trunk length and cumulative scaled reproductive energy to dry weight. If dry weight data are available as well, there are seven parameters to be estimated from three curves.

Figure 2.19 gives an example. The data appear to contain too little information to determine both $\dot{k}_M$ and g, so either $\dot{k}_M$ or g has to be fixed. The more or less arbitrary choice $g = 0.4$ is made here. The estimates are tied by the relationship that $\frac{\dot{k}_M g}{1+g}$ is almost constant. The high value for the maintenance rate coefficient $\dot{k}_M$ probably relates to the investment of energy in the frequent synthesis of new filtering houses. The cumulative reproduction fits the data for *D. magna* also very well (see Figure 2.10), where $\dot{k}_M$ is even higher, probably related to moulting.

2.7.2 Buffer handling rules

Individuals are discrete units, which implies the existence of a buffer, where the energy allocated to reproduction is accumulated and converted to eggs at the moment of repro-

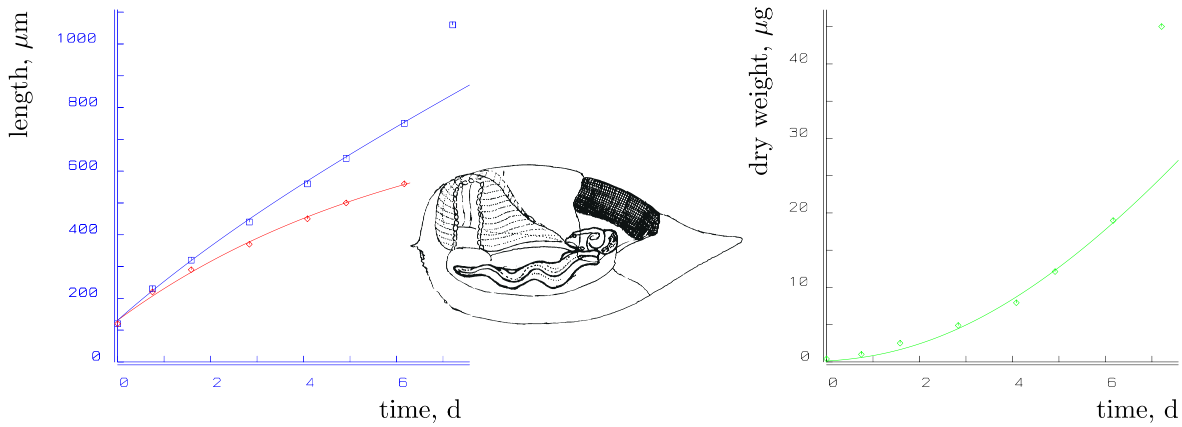


Figure 2.19: The total trunk length, L_t ($\square$ and upper curve, left), the true trunk length, L ($\diamond$ and lower curve, left) and the dry weight (right) for *Oikopleura longicauda* at 20 °C. Data from Fenaux and Gorsky [341]. The DEB-based curves account for the contribution of the accumulated energy, allocated to reproduction, to total trunk length and to dry weight. The parameter estimates are $L_m = 822\,\mu\text{m}$, $l_b = l_p = 0.157$, $\dot{k}_M = 1.64\,\text{d}^{-1}$, $g = 0.4$, $V_R\kappa_R = 0.0379\,\text{mm}^3$. Given these parameters, the weight data give $W_{Ld} = 0.0543\,\text{g}\,\text{cm}^{-3}$, $W_{Rd} = 15.2\,\mu\text{g}$. The last data point is excluded in both data sets because here structure is rapidly converted to gametes, and maintenance has probably ceased. This 'detail' is not implemented in the standard DEB model.

duction. The translation of reproduction rate into number of eggs in Figure 2.17 assumes that this accumulation is over a period of one year. The energy content of the buffer is denoted by E_R.

The strategies for handling this buffer are species-specific. Some species (e.g. some rotifers) reproduce when enough energy for a single egg has been accumulated, others wait longer and produce a large clutch. If the reproduction buffer is used completely, the size of the clutch equals the ratio of the buffer content to the energy costs of one young, $\kappa_R E_R/E_0$, where E_0 is given in (2.42). This resets the buffer. So after reproduction $E_R = 0$ and further accumulation continues from there. That is to say, the bit of energy that was not sufficient to build the last egg can become lost or still remains in the buffer; fractional eggs do not exist. In the chapter on population dynamics, {344,346}, I show that this uninteresting detail substantially affects dynamics at low population growth rates, which occur most frequently in nature. If food is abundant, the population will evolve rapidly to a situation in which food per individual is sparse and reproduction low if harvesting processes do not prevent this.

Reproduction is coupled to the moulting cycle in daphnids; neonates in the brood pouch are released, the old moult is shedded, the new eggs are deposited in the brood pouch and the new carapace hardens out and the cycle repeats some 1.5 to 2 days later at 20 °C. The moulting cycle is linked to somatic maintenance and, therefore, is independent of nutritional status, while the incubation time is. This means that there must be a variable diapause.

Many species use environmental triggers for spawning at particular times t_R. Many molluscs spawn if temperature exceeds a threshold value, while $[E_R]$ is larger than a threshold value [917]. Other species use food availability as trigger; *Pecten maximus* spawns just after an algal bloom. The spectacular synchronisation of reproduction in corals [1049], the pelagic palolo worms *Eunice viridis*, South East Asian dipterocarps and bamboo forests probably reduce losses, because potential predators have little to eat between the events. Most species are able to synchronise the moment of reproduction with seasonal cycles such that food availability just matches the demand of the offspring. Clutch size in birds typically relates to food supply during a 2-month period prior to egg laying and tends to decrease if breeding is postponed in the season [778]. The laying date is determined by a rapid increase in food supply. Since feeding conditions tend to improve during the season, internal factors must contribute to the regulation of clutch size. These conclusions result from an extensive study of the energetics of the kestrel *Falco tinnunculus* by Serge Daan and co-workers [269, 753, 777]. I see reproductive behaviour like this for species that cease growth at an early moment in their life span, as variations on the general pattern that the DEB theory is aiming to grasp. Aspects of reproduction energetics for species that cease growth are worked out on {277}.

Multiple spawning

Batch preparation in adult anchovy is initiated when the temperature in spring exceeds a threshold (14 °C in anchovy [875, 876]); juveniles that mature after this time point have to wait for batch preparation till the next spring. The batch size expressed to total amount of reserves is proportional to structural volume; division by the cost per egg gives the number of eggs per spawning, but this involves the reserve density of the spawner. This means that if the scaled functional response decreases during the spawning season, the numbers of eggs increases (if length remains constant).

The rate of batch preparation equals the maximum allocation to reproduction (i.e. as if $e = 1$) and batch preparation ceases for that spawning season if the reproduction buffer is emptied. The rate still depends on length of the individual and is motivated by the avoidance of an unbounded accumulation of the reproduction buffer with abundant food (during the whole year). The spawning season, however, lasts less than a year, so the rate of batch preparation is divided by the fraction of the year that has good spawning conditions, which is about 7/12. Only in the last batch of the spawning season will the batch size be smaller than the target size. If food were abundant, this rule for spawning would imply that spawning, once initiated, continues till death.

2.7.3 Post-reproductive period

Many animal species have a post-reproductive period. In the context of DEB theory, this is (or can be) implemented as an aspect of ageing, see {209}, which is taken to be an effect of reactive oxygen species (ROS). Effects of compounds are, in general, implemented as affects on particular DEB parameters, like effects of temperature. This implementation requires more state variables than the standard DEB model has.

2.8 Parameter estimation I: numbers, lengths and time

Parameters with energy or moles in their dimension require measurements of energy and moles, respectively. The compound parameter g is dimensionless, and $\dot{k}_M$ has only time in its dimension, so they don't need such measurements to be estimated. The DEB parameters that can be extracted from observations on length and reproduction over time at several food densities are: κ, g, $\dot{k}_J$, $\dot{k}_M$, $\dot{v}$, $U_H^b = E_H^b/\{\dot{p}_{Am}\}$ and $U_H^p = E_H^p/\{\dot{p}_{Am}\}$, see [660] and Figure 2.10. These parameters also determine the initial scaled reserve $U_E^0 = E_0/\{\dot{p}_{Am}\}$ as a function of scaled functional response f. A data set using a single measured time-varying food density would be enough, in principle, but the estimation process is more complex, compared to a set (> 1) of constant food densities. The food densities only have to be measured in the latter case if $\{\dot{F}_m\}$ needs to be obtained. The parameter κ_R must be obtained from mass balances; a default value of $\kappa_R = 0.95$ would probably be appropriate in most cases.

Common practice is that these observations are not always available, and the question is: what can be done with fewer data? If observations on just a single food density are available, we don't know how size at birth and/or puberty depends on food availability, and we are forced to assume that $\dot{k}_J = \dot{k}_M$.

The (compound) DEB parameters that only have time and length in their dimension can be obtained from growth and reproduction data with functions in DEBtool. These parameters don't depend on food level, while the growth and reproduction data do. To emphasise this, the quantities that depend on food level are printed bold below:

Growth at a single food level: debtool/animal/get_pars_g

$$(L_b, \boldsymbol{L}_\infty, \boldsymbol{a}_b, \dot{\boldsymbol{r}}_B \text{ at } \boldsymbol{f}_1) \longrightarrow (g, \dot{k}_M = \dot{k}_J, \dot{v}; \boldsymbol{U}_E^0, \boldsymbol{U}_E^b \text{ at } \boldsymbol{f}_1)$$

Growth at several food levels: debtool/animal/get_pars_h

$$\begin{pmatrix} \boldsymbol{L}_b, \boldsymbol{L}_\infty, \dot{\boldsymbol{r}}_B \text{ at } \boldsymbol{f}_1 \\ \boldsymbol{L}_b, \boldsymbol{L}_\infty, \dot{\boldsymbol{r}}_B \text{ at } \boldsymbol{f}_2 \end{pmatrix} \longrightarrow \begin{pmatrix} g, \dot{k}_M, \dot{k}_J, \dot{v}, & \begin{matrix} \boldsymbol{U}_E^0, \boldsymbol{U}_E^b \text{ at } \boldsymbol{f}_1 \\ \boldsymbol{U}_E^0, \boldsymbol{U}_E^b \text{ at } \boldsymbol{f}_2 \end{matrix} \end{pmatrix}$$

Growth at several food levels: debtool/animal/get_pars_i

$$\begin{pmatrix} L_b, & \begin{matrix} \boldsymbol{L}_\infty, \dot{\boldsymbol{r}}_B \text{ at } \boldsymbol{f}_1 \\ \boldsymbol{L}_\infty, \dot{\boldsymbol{r}}_B \text{ at } \boldsymbol{f}_2 \end{matrix} \end{pmatrix} \longrightarrow \begin{pmatrix} g, \dot{k}_M = \dot{k}_J, \dot{v}, & \begin{matrix} \boldsymbol{U}_E^0, \boldsymbol{U}_E^b \text{ at } \boldsymbol{f}_1 \\ \boldsymbol{U}_E^0, \boldsymbol{U}_E^b \text{ at } \boldsymbol{f}_2 \end{matrix} \end{pmatrix}$$

Growth and reproduction at a single food level: debtool/animal/get_pars_r

$$(L_b, L_p, \boldsymbol{L}_\infty, \boldsymbol{a}_b, \dot{\boldsymbol{r}}_B, \dot{\boldsymbol{R}}_\infty \text{ at } \boldsymbol{f}_1) \overset{\text{given } \kappa_R}{\longrightarrow} (\kappa, g, \dot{k}_J = \dot{k}_M, \dot{v}, U_H^b, U_H^p; \boldsymbol{U}_E^0, \boldsymbol{U}_E^b, \boldsymbol{U}_E^p \text{ at } \boldsymbol{f}_1)$$

Growth and reproduction at several food levels: debtool/animal/get_pars_s

$$\begin{pmatrix} \boldsymbol{L}_b, \boldsymbol{L}_p, \boldsymbol{L}_\infty, \dot{\boldsymbol{r}}_B, \dot{\boldsymbol{R}}_\infty \text{ at } \boldsymbol{f}_1 \\ \boldsymbol{L}_b, \boldsymbol{L}_p, \boldsymbol{L}_\infty, \dot{\boldsymbol{r}}_B, \dot{\boldsymbol{R}}_\infty \text{ at } \boldsymbol{f}_2 \end{pmatrix} \overset{\text{given } \kappa_R}{\longrightarrow} \begin{pmatrix} \kappa, g, \dot{k}_J, \dot{k}_M, \dot{v}, U_H^b, U_H^p, & \begin{matrix} \boldsymbol{U}_E^0, \boldsymbol{U}_E^b, \boldsymbol{U}_E^p \text{ at } \boldsymbol{f}_1 \\ \boldsymbol{U}_E^0, \boldsymbol{U}_E^b, \boldsymbol{U}_E^p \text{ at } \boldsymbol{f}_2 \end{matrix} \end{pmatrix}$$

Growth and reproduction at several food levels: debtool/animal/get_pars_t

$$\begin{pmatrix} L_b, L_p, & \boldsymbol{L}_\infty, \dot{\boldsymbol{r}}_B, \dot{\boldsymbol{R}}_\infty \text{ at } \boldsymbol{f}_1 \\ & \boldsymbol{L}_\infty, \dot{\boldsymbol{r}}_B, \dot{\boldsymbol{R}}_\infty \text{ at } \boldsymbol{f}_2 \end{pmatrix} \overset{\text{given } \kappa_R}{\longrightarrow} \begin{pmatrix} \kappa, g, \dot{k}_J = \dot{k}_M, \dot{v}, U_H^b, U_H^p, & \boldsymbol{U}_E^0, \boldsymbol{U}_E^b, \boldsymbol{U}_E^p \text{ at } \boldsymbol{f}_1 \\ & \boldsymbol{U}_E^0, \boldsymbol{U}_E^b, \boldsymbol{U}_E^p \text{ at } \boldsymbol{f}_2 \end{pmatrix}$$

The unscaled reserve E and maturity E_H require energies, and thus knowledge of $\{\dot{p}_{Am}\}$. This can be obtained from observations on the feeding process and the mass at zero and birth. These extra observations give access to more parameters, and are discussed at {163}. DEBtool also has functions iget_pars that do the inverse mapping from (compound) DEB parameters to easy-to-measure quantities. This can be used for checking the mapping and testing against empirical data.

2.9 Summary of the standard DEB model

The standard DEB model applies to an isomorph that feeds on a single type of food and has a single reserve and a single structure. In this chapter we have used time, length and energy only; this is not always most convenient. The model has three state variables: structural volume V, reserve energy E and maturity, expressed in terms of cumulative energy investment, E_H. Maturity has no mass or energy, however, and represents information. The input variable is food density $X(t)$, and temperature $T(t)$ modifies all rates (which can be recognised by the dots); they typically vary in time in harmony. The model has 12 individual-specific parameters: specific searching rate $\{\dot{F}_m\}$, assimilation efficiency κ_X, maximum specific assimilation rate $\{\dot{p}_{Am}\}$, energy conductance $\dot{v}$, allocation fraction to soma κ, reproduction efficiency κ_R, volume-specific somatic maintenance cost $[\dot{p}_M]$, surface area-specific somatic maintenance cost $\{\dot{p}_T\}$, maturity maintenance rate coefficient $\dot{k}_J$, specific cost for structure $[E_G]$, maturity at birth E_H^b and maturity at puberty E_H^p. Typical parameter values are discussed in Chapter 8.

The 10 assumptions that fully specify the standard DEB model are listed in Table 2.4. If the somatic and maturity rate coefficients are equal, birth and puberty occur at fixed amounts of structure, so that there is no longer a need for maturity as an explicit state variable; otherwise the scaled length at birth l_b and l_p are not constant. Other DEB models are modified versions of the standard DEB model to include dividing organisms, changing of shapes, multiple types of food, reserve and structure, adaptation, etc. Effects of compounds, such as ageing, need more state variables.

Table 2.5 gives the resulting energy fluxes, called powers, as functions of the scaled reserve density e and scaled length l, where the scaled length at birth l_b and puberty l_p depend on the food history if $\dot{k}_J \neq \dot{k}_M$. The relationships between compound and primary parameters are summarised in Table 3.3. Notice that all powers are cubic polynomials in the (scaled) length, while the weight coefficients depend on (scaled) reserve density.

The mobilisation power equals the sum of the non-assimilative powers, and κ times the mobilisation power equals the sum of somatic maintenance and growth:

$$\dot{p}_C = \dot{p}_S + \dot{p}_G + \dot{p}_J + \dot{p}_R \quad \text{with } \kappa\dot{p}_C = \dot{p}_S + \dot{p}_G \tag{2.60}$$

Table 2.4: The assumptions that specify the standard DEB model quantitatively.

1 The amounts of reserve, structure and maturity are the state variables of the individual; reserve and structure have a constant composition (strong homeostasis) and maturity represents information.

2 Substrate (food) uptake is initiated (birth) and allocation to maturity is redirected to reproduction (puberty) if maturity reaches certain threshold values.

3 Food is converted into reserve and reserve is mobilised at a rate that depends on the state variables only to fuel all other metabolic processes.

4 The embryonic stage has initially a negligibly small amount of structure and maturity (but a substantial amount of reserve). The reserve density at birth equals that of the mother at egg formation (maternal effect). Foetuses develop in the same way as embryos in eggs, but at a rate unrestricted by reserve availability.

5 The feeding rate is proportional to the surface area of the individual and the food–handling time is independent of food density.

6 The reserve density *at constant food density* does not depend on the amount of structure (weak homeostasis).

7 Somatic maintenance is proportional to structural volume, but some components (osmosis in aquatic organisms, heating in endotherms) are proportional to structural surface area.

8 Maturity maintenance is proportional to the level of maturity

9 A fixed fraction of mobilised reserves is allocated to somatic maintenance plus growth, the rest to maturity maintenance plus maturation or reproduction (the κ-rule).

10 The individual does not change in shape during growth (isomorphism). This assumption applies to the standard DEB model only.

A three-stage individual invests either in maturation, or in reproduction. This is why these powers have the same index; the stage determines the destination.

The dissipating power excludes assimilation and somatic growth overheads by definition and amounts to

$$\dot{p}_D = \dot{p}_S + \dot{p}_J + (1 - \kappa_R)\dot{p}_R \tag{2.61}$$

where $\kappa_R = 0$ for the embryo and juvenile stages. Reproduction power $\dot{p}_R$ has a special status because reserve of the adult female are converted into reserve of the embryo(s) which have the same composition; $(1 - \kappa_R)\dot{p}_R$ is dissipating and $\kappa_R \dot{p}_R$ returns to the reserve, but now of the embryo.

Table 2.5: The scaled powers $\dot{p}_*/\{\dot{p}_{Am}\}L_m^2$ as specified by the standard DEB model for an isomorph of scaled length $l = L/L_m$, scaled reserve density $e = [E]/[E_m]$ and scaled maturity density $u_H = \frac{E_H}{g[E_m]V_m}$ at scaled functional response $f \equiv \frac{X}{K+X}$, where X denotes the food density and K the saturation constant. The powers $\dot{p}_X$ and $\dot{p}_P$ for ingestion and defecation occur in the environment, not in the individual. Assimilation is switched on at scaled length l_b, and allocation to maturation is redirected to reproduction at $l = l_p$. Compound parameters: allocation fraction κ, investment ratio g, somatic maintenance rate coefficient $\dot{k}_M$, scaled heating length l_T, maintenance ratio k. Implied dynamics for $e > l > l_b$:
$\frac{d}{dt}e = \dot{k}_M g \frac{f-e}{l}$ and $\frac{d}{dt}l = \frac{\dot{k}_M}{3}\frac{e-l-l_T}{1+e/g}$ and $\frac{d}{dt}u_H = \dot{k}_M(u_H < u_H^p)((1-\kappa)el^2\frac{g+l+l_T}{g+e} - ku_H)$.

$\frac{\text{power}}{\{\dot{p}_{Am}\}L_m^2}$	embryo $0 < l \leq l_b$	juvenile $l_b < l \leq l_p$	adult $l_p < l < 1$
assimilation, $\dot{p}_A$	0	fl^2	fl^2
mobilisation, $\dot{p}_C$	$el^2\frac{g+l}{g+e}$	$el^2\frac{g+l+l_T}{g+e}$	$el^2\frac{g+l+l_T}{g+e}$
somatic maintenance, $\dot{p}_S$	κl^3	$\kappa l^2(l+l_T)$	$\kappa l^2(l+l_T)$
maturity maintenance, $\dot{p}_J$	ku_H	ku_H	ku_H^p
growth, $\dot{p}_G$	$\kappa l^2\frac{e-l}{1+e/g}$	$\kappa l^2\frac{e-l-l_T}{1+e/g}$	$\kappa l^2\frac{e-l-l_T}{1+e/g}$
maturation, $\dot{p}_R$	$(1-\kappa)el^2\frac{g+l}{g+e} - ku_H$	$(1-\kappa)el^2\frac{g+l+l_T}{g+e} - ku_H$	0
reproduction, $\dot{p}_R$	0	0	$(1-\kappa)el^2\frac{g+l+l_T}{g+e} - ku_H^p$

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